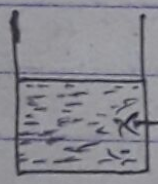
THERMODYNAMICS

→ deal with various form of energy and their interconversion in physical or chemical changes, it also helps in predicting feasibility of a process or reaction under given conditions.

* Basic terms used in thermodynamics :

System : (Part of universe under thermodynamic study)

(i) Open System :
where energy (heat or work) and matter both can exchange between system and surroundings.

eg. 

open vessel having water.
evaporation will take place, temperature may change

(ii) Closed System :
only energy exchange occurs.

eg.

Closed vessel.

(iii) Isolated System :
Neither energy nor matter exchange.
eg. Thermos flask, Universe.
↳ Perfect isolated system.

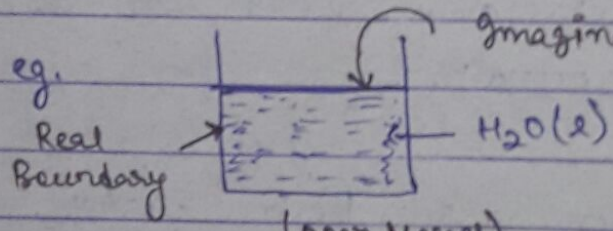
⊛ Surroundings :
Part of the universe other than system.

⊛ Boundary :
Separation between system and surroundings.

Classification of Boundaries:

- Rigid & non - rigid boundary.
  - Rigid → $\Delta V = 0$ (Thick walled vessel)
  - non-rigid → $\Delta V \neq 0$ (eg. balloon)

- Real and imaginary boundary



imaginary boundary. (liquid surface).

- Adiabatic and diathermic

↑ the wall which does not allow any heat transfer across it. (i.e $q = 0$) → made of insulating material. eg. Thermos flask.

↖ which allows heat transfer. → made of conducting material. eg. Copper vessel.

* Intensive Properties :-

Property whose value remain unchanged when system is divided into smaller parts.

eg. Pressure, temperature, density, specific heat, Molar heat capacity, EMF of cell, surface tension, Viscosity, pH of solution, Refractive index, Molarity & concentration terms, specific volume etc.

* Extensive Properties :-

Property whose value is changed when system is divided into smaller parts.

eg. mass, volume, Heat capacity, Enthalapy (H), Entropy (S), Internal energy (U or E), Gibb's Energy (G) etc.

Note:

Extensive property per gram or per mol becomes intensive property.

eg. Enthaly → extensive
Molar enthalpy → intensive

Mass → extensive property
Molar → intensive property

Heat capacity → extensive
Specific Heat
Molar heat capacity
] intensive.

Molar heat capacity)

⇒ Molar heat capacity = specific heat × Molar mass

J/mol/°C J/g/°C

(*) State function (or State Variable) (ϕ)

→ Certain measurable properties are required to describe the state of a thermodynamic system, called state variables.

eg- Pressure, temperature, Volume, Enthalpy, Entropy, Gibb's energy, Internal energy.

→ State function depends only on current state of system and does not depend on previous state or path followed to attain the current state.

Imp. → Change in state function between two states does not depend on path followed.

ϕ → state function,

$$\Delta\phi = \int_1^2 d\phi = \phi_2 - \phi_1$$

∴ $\Delta T = T_2 - T_1$

$\Delta P = P_2 - P_1$

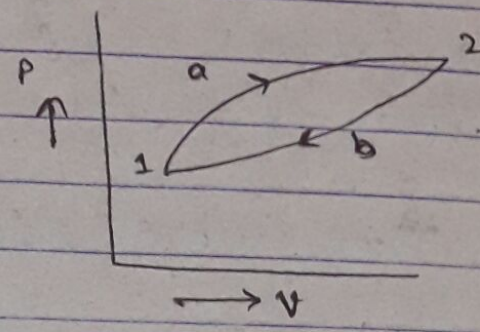
$\Delta S = S_2 - S_1$ etc.

* Change in state function between two states is a definite quantity.

* Change in state function is not a state function.
⇒ ΔT is not a state function.

temperature is a state fn. but change in temperature is a path fn.

→ Change in State function in a cyclic process is equal to zero.



cyclic integral

$$\oint d\phi = \int_1^2 d\phi + \int_2^1 d\phi$$

↓

Change of state function in cyclic process

$$= (\phi_2 - \phi_1) + (\phi_1 - \phi_2)$$
$$= 0$$

$\Rightarrow \Delta T = 0, \quad \Delta S = 0, \quad \Delta P = 0, \quad \Delta \phi = 0 \ldots\ldots$

in a cyclic process

→ $\phi = f(x, y)$

$$d\phi = \left(\frac{\partial \phi}{\partial x}\right)_y dx + \left(\frac{\partial \phi}{\partial y}\right)_x dy$$

Rate of change of ϕ w.r.t change in x.

x, y → independent variables.

ϕ → state function depending upon x and y.

Change in $x \rightarrow dx$

Change in $y \rightarrow dy$

$z = f(x)$

$\frac{dz}{dx} = f'(x)$

But if

$z = f(x, y)$

$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right) dy$

∂ → read as

Eg. Ideal gas, $PV = nRT$, for a fixed amount of gas,

$P = f(T, V)$ (or) $T = f(P, V)$ (or) $V = f(T, P)$

$$dP = \left(\frac{\partial P}{\partial T}\right)_V dT + \left(\frac{\partial P}{\partial V}\right)_T dV$$

Note :

Path function :
Property of path i.e depends on path followed.

eg. Heat, work, Molar Heat Capacity.

(*) Process :-
A path through which change of state of system occurs.

(i) Isothermal process :
Temperature constant throughout, $dT = 0$ & $\Delta T = 0$

(ii) Isochoric process :
Volume constant throughout, $dV = 0$, $\Delta V = 0$

(iii) Isobaric process :
Pressure constant throughout, $dp = 0$, $\Delta P = 0$

(iv) Adiabatic process :
No heat transfer between system & surroundings.
$(q = 0)$

(v) Cyclic process :
When system undergoes a series of processes and finally returns to the original state.

$\Delta \phi = 0$

$\Rightarrow \Delta T = 0$, $\Delta P = 0$, $\Delta S = 0$, $\Delta H = 0$ etc.

* Reversible process : (Quasi static process)

infinitesimally slow process

→ A ~~proce~~ reversible process is an infinitesimally slow process and takes infinite no of steps for a finite change. This occurs when pressure difference and/or temp. difference between system and surroundings are infinitesimally small.

$$\left.\begin{aligned} P_{ext} &= P \pm dp \\ T_{ext} &= T \pm dT \end{aligned}\right\}$$

→ In a reversible process, both system and surroundings are in thermodynamic equilibrium at every instant during a process.

→ System and Surroundings are said to be in thermodynamic equilibrium when,

(i) Mechanical equilibrium ($\Delta P = 0$)

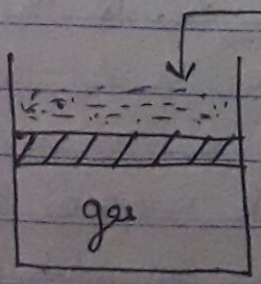
(ii) Thermal equilibrium ($\Delta T = 0$)

(iii) Chemical equilibrium, exists between System & surroundings.

there is uniform composition of the system and no net chemical rxn. is occuring

→ Reversible process is a hypothetical process and can't be attained in practice.

eg.



→ When path of reversible process is reversed, then both system and surroundings regain their original state.

(*) Irreversible process:
Unlike reversible process, irreversible process completes in a definite no. of steps.

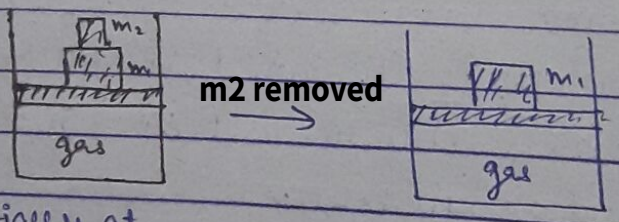
→ All natural process are irreversible processes.

→ There is a finite difference in pressure & temperature of system and surroundings.

$$P_{ext} = P \pm \Delta P$$
$$T_{ext} = T \pm \Delta T$$

these finite differences drive the process to move relatively fast

→ eg.



Initially at equilibrium

Finally at eqm.

Irreversible expansion.

Imp → In irreversible process, thermodynamic eqm. exists only at initial and final states.

there is no thermodynamic equilibrium during the process

→ When path of irreversible process is reversed, either system or surrounding or both do not get back to their original States.

* Modes of energy transfer between System and Surroundings :-

1. Work : (Path function)

- PV work (or) mechanical work (or) displacement work) due to volume change.

- non-PV work (non-mechanical work)
eg - (Electrical work)

IUPAC Sign Convention :

In Chemistry, any energy given to system is taken +ve.

→ Work done on system (work of compression) = +ve

→ work done by the system (or work of expansion) = -ve

Note : eg. work done = -50 KJ

OR

work done by the system = 50 J

Sy eg. work done = +100 KJ

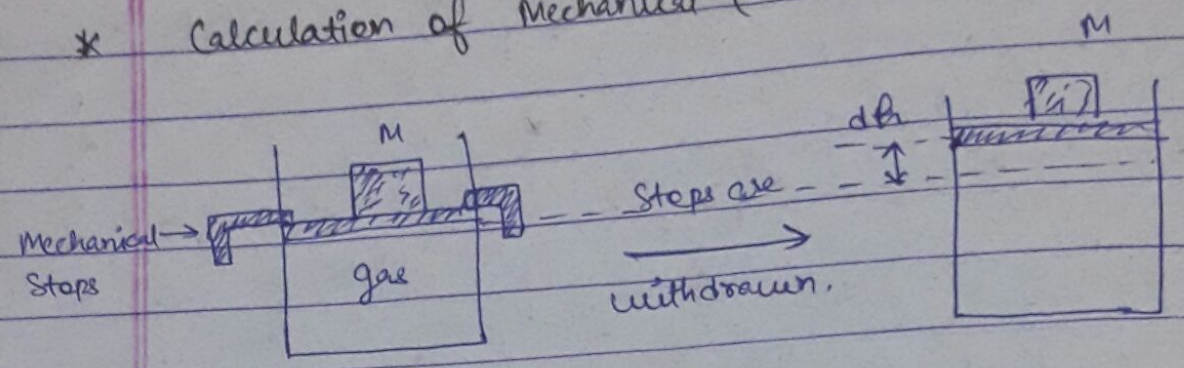
OR

work done on the system = 100 KJ

OR

work done by the system = -100 KJ

* Calculation of Mechanical (or PV work)



Initially $P_{gas} > P_{ext}$

Magnitude of work done = Change in P.E of mass M.

$$= \frac{Mg}{A} \times dh \times A$$

$$= P_{ext} \times dV$$

∵ here expansion has occured so work done will be -ve.

Work done = $-P_{ext}\, dV$

Total work done =

$$W = -\int_1^2 P_{ext} \times dV$$

⇒ If ext. pressure is constant, then

$$W = -P_{ext.} \times \Delta V$$

→ Work done (or PV work) will be zero when
(i) Process is isochoric i.e $dV = 0$ & $\Delta V = 0$
(ii) Process is free expansion (or expansion against vacuum) i.e $P_{ext} = 0$.
{ Adiabatic free expansion }

(2) Heat (q or Q) :

It flows between System and surroundings by virtue of temperature difference.

Heat given to System = +ve

Heat taken from system = - ve

* Molar Heat Capacity (c) :

It is the heat required to raise temperature of 1 mol of substance through one degree.

$$c = \frac{q}{dT}$$ where q is heat required to raise temp. of 1 mol of substance by dT.

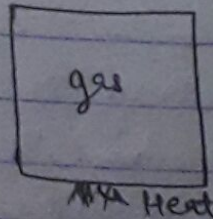
$q = cdT$ for 1 mol

$q = nCdT$ for n mol

$$Q = n\int_1^2 CdT = nC\Delta T \text{ if 'C' is constant.}$$

* Process dependent Molar Heat Capacity :—

At constant volume (C_v or $C_{v,m}$)



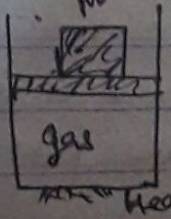
$q_v = nC_vdT$

$Q_v = \int_1^2 nC_vdT$

$Q_v = nC_v\Delta T$

At constant pressure (C_p or $C_{p,m}$)

piston type vessel



$q_p = nC_pdT$

$Q_p = n\int_1^2 C_pdT$

$Q_p = nC_p\Delta T$

valid when Cp and Cv are temperature independent

Note: (1) for a gas, $C_p > C_v$

→ for an ideal gas, $\boxed{C_p - C_v = R}$

as rotational & vibrational DOF are absent

Imp: (2) Except monoatomic gas (He, Ne, Ar, etc) C_p or C_v or C for every substance (including ideal gas) depend upon temperature.

even for solids and liquids

as rotational & vibrational DOF are temperature dependent

In general, $C = a + bT + CT^2 + \cdots\cdots$

(3) For an ideal gas, (treating the gases ideal)

Poisson's ratio

atomicity	C_v	C_p	$\gamma = C_p/C_v$
• monoatomic gas (He, Ne, Ar)	$3/2\ R$	$5/2\ R$	$5/3 \approx 1.66$
• Diatomic linear & polyatomic gas ($O_2, H_2, N_2, N_2O, CO_2$)	$5/2\ R$	$7/2\ R$	$7/5 = 1.4$
• Polyatomic non-linear gases. (O_3, H_2O, SO_2, CH_4 etc)	$3R$	$4R$	$4/3 \approx 1.33$

⇒ All these C_p, C_v values are taken, considering only translation and Rotational degree of freedom. Here Vibrational degree of freedom are not considered as they become active at very high temperature.

→ Molar Heat Capacity = Specific Heat × Molar Mass

↑ J/mol/K ↑ J/g/K

(*) Internal Energy (U or E) :
(state function)

Every thermodynamic system at rest and in absence of any external force field (EF or MF) possess a certain amount of energy associated with its molecules, called internal energy.

* Internal energy of a system includes

→ K.E (Transe. + Rot + Vib)
→ P.E (due to intermolecular forces) = 0 for ideal gas
→ ~~B.E~~ Bond Energies
→ Electronic Energies .....

whenever internal energy changes, changes occur only in KE and PE. Bond energies and electronic energies don't change unless a chemical rxn. is taking place.

→ Absolute value of internal energy can't be determined. infact not needed. We need change in internal energy when system goes from one state to another.

$$\Delta U = \Delta(K.E) + \Delta(P.E)$$
$$K.E = f(T), \quad P.E = f(T, V)$$

⇒ In general, we can say that,
$$U = f(T, V) \text{ or } f(T, P)$$

Imp. → For an ideal gas, $P.E = 0$, $\Delta(P.E) = 0$

$\Delta U = \Delta(K.E)$

$U = f(T)$ for an ideal gas.

In general, $U = f(T, V)$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT.$$

$$\Rightarrow dU = C_V dT + \left(\frac{\partial U}{\partial V}\right)_T V \quad \text{—— (1)}$$

Imp. Case - I for an ideal gas undergoing any process,

$U = f(T)$

$$\left(\frac{\partial U}{\partial V}\right)_T = 0$$

$\Rightarrow dU = C_V dT$ for 1 mol

$dU = nC_V dT$ for n mol

$$\Delta U = n\int_1^2 C_V dT$$

$$\Delta U = nC_V \Delta T$$

⊙ for an ideal gas

⊙ for every process.

Case - II for a real gas undergoing isochoric process.

$V \rightarrow$ constant, $dV = 0$

$$dU = C_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV \quad \text{(zero)}$$

$$\boxed{\Delta U = n \int_1^2 C_V dT}$$ at isochoric process.

incompressible phases

Case - III for solids & liquids undergoing any process

$dV \approx 0$, $C_p \approx C_V \cong C$

$$\Delta U = n \int_1^2 C_V dT + 0$$

$$\boxed{\Delta U = n \int_1^2 C dT}$$

(*) 1st law of thermodynamics :-

It is another form of law of conservation of energy, according to which energy can neither be created nor destroyed. (OR)

Imp Total energy of universe is constant.

Mathematically,

$$U_2 = U_1 + W + Q$$

$$\boxed{\Delta U = Q + W}$$ → applicable for closed system.

$$\boxed{dU = q + W}$$

Q: Calculate amount of heat required to raise temperature of 2 mol of a gas from 300k to 400k if C of gas is given by, $C = (20 + 10^{-3}T)$ J/mol/k.

Soln:

$$Q = n\int_1^2 C\,dT$$

$$= 2\int_1^2 (20 + 10^{-3}T)\cdot dT$$

$$= 2\left[20 \times (400-300) + \frac{10^{-3}}{2}\left[(400)^2 - (300)^2\right]\right]$$

$$= 2[2000 + 5 \times (16-9)]$$

$$= 2[2000 + 35]$$

$$= 4070 \text{ Joule}$$

Q. 200 ml of liquid water is compressed by a pressure of 100 bar so that 2% compression of initial volume occurs. calculate the work done.

Soln: $$\text{work done} = -\int_1^2 P_{ext} \times dV$$

$$= -P_{ext} \times \Delta V$$

$$= -100 \times 10^5 \text{ Pa} \times (-4) \times 10^{-6} \text{ m}^3$$

$$= 40 \text{ Joule.}$$

Q. 16 g of O_2 gas (Perfect or ideal gas) is given a heat of 100 Joule and temp. rise is found to be 20°C. Find the work done.

Soln: $$\Delta U = Q + W$$

$$\Delta U = nC_v \Delta T$$

$$= \frac{16}{32} \times \frac{5}{2} R \times 20$$

$$= \frac{5}{4} \times \frac{25}{3} \times 20 \text{ Joule}$$

$$= \frac{2500}{12} = 208.3 \text{ Joule.} = 208.3 \text{ Joule}$$

$$W = \Delta U - Q$$

$$= 208.3 - 100$$

$$= 108.3 \text{ Joule}$$

* Enthalpy (H) : (State function)

$$H = U + PV$$

$$dH = dU + d(PV)$$

$$dH = dU + PdV + VdP$$

At constant pressure, $dP = 0$, $q = q_p$

$$dH = dU + PdV \quad \text{——(i)}$$

From 1st law of Thermodynamics,

$$dU = q + W$$

$$dU = q_p - PdV$$

$$dH = q_p - PdV + PdV$$

$$\boxed{dH = q_p}$$

$$dU = q + W$$

At constant volume,

$$q = q_v, \quad W = -PdV = 0$$

$$\boxed{dV = q_v}$$

→ Considering only PV work,

V. imp

Heat given at constant vol. = Change in internal energy

(or)

$$\boxed{Q_v = \Delta U}$$

⇒ Heat given at constant pressure = Change in enthalpy

$$\boxed{Q_p = \Delta H}$$

$$U = f(T, V)$$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

$$\boxed{dU = C_v dT + \left(\frac{\partial U}{\partial V}\right)_T dV}$$

absolute value of enthalpy also cannot be calculated

⇒ Expression of enthalpy change :

for an ideal gas, $H = U + PV$

$$H = U + nRT$$

$$U = f(T)$$

⇒ $H = f(T)$, only for a ideal gas.

⇒ But for a real gas or solid or liquid or in general.

$$H = f(T, P)$$

$$dH = \left(\frac{\partial H}{\partial T}\right)_P dT + \left(\frac{\partial H}{\partial P}\right)_T dP$$

$$dH = q_p \cdot \Rightarrow \left(\frac{\partial H}{\partial T}\right)_P = C_p$$

$$\therefore \boxed{dH = C_p dT + \left(\frac{\partial H}{\partial P}\right)_T dP} \quad \text{---(ii)}$$

Case - I for an ideal gas undergoing any process,

$$H = f(T), \quad \left(\frac{\partial H}{\partial P}\right)_T = 0$$

$\Rightarrow dH = C_p dT$ for 1 mol

$dH = nC_p dT$ for n mol

$$\Rightarrow \boxed{\Delta H = n\int_1^2 C_p dT}$$

$$\boxed{\Delta H = nC_p \Delta T}$$ → for an ideal gas, undergoing any process.

Case - II for a real gas undergoing isobaric process,

$\Rightarrow P \rightarrow$ constant, $\left\{\frac{\partial H}{\partial P}\right\}_T dp = 0$

$$\boxed{\Delta H = nC_p \Delta T}$$

Case - III for solid or liquid, ~~Effect of pressure~~ is negligible

$$\left(\frac{\partial H}{\partial P}\right)_T \approx 0$$

$\Rightarrow dH = nCdT$

$\Delta H = n\int_1^2 CdT$ $\quad [C_p \equiv C_v \equiv C]$

$$\boxed{\Delta H = nC\Delta T}$$ if 'C' is constant,

* Relation between ΔH and ΔU :

$$H = U + PV$$

$$\Delta H = \Delta U + \Delta(PV)$$

$$\boxed{\Delta H = \Delta U + (P_2V_2 - P_1V_1)} \quad \text{---} \quad (1)$$

→ If P & V data at initial and final states are known, then above equation (1) can be applied.

→ If data is insufficient, then suitable approximations can be done.

(1) for an ideal gas undergoing any process :

$$\Delta H = \Delta U + \Delta(PV) \quad , \Rightarrow PV = nRT$$

$$\Delta H = \Delta U + \Delta(nRT)$$

$$\boxed{\Delta H = \Delta U + nR\Delta T}$$

For an isothermal process, for ideal gas,

$$\left.\begin{array}{l} \Delta H = 0 \\ \Delta U = 0 \end{array}\right\} \quad \begin{array}{l} U = f(T) \\ H = f(T) \end{array}$$

(2) for a chemical reaction or phase change occuring at constant temperature :

$$\Delta H = \Delta U + \Delta(PV)$$

→ For a rxn. involving gas (gases), $PV = nRT$
(gases are ideal)

$$\Delta(PV) = \Delta(nRT)$$
$$= \Delta n_g \times RT$$

$\therefore$ $\Delta(PV)$ is ignored for solids and liquids in comparison to gases.

$\Rightarrow$ $\boxed{\Delta H = \Delta U + \Delta n_g \times RT}$ —— (iii)

↳ (i) for a rxn / phase change
(ii) Gas are treated ideal.

Note: If gases involved in the reaction are ~~treated~~ to be real (or mentioned to be deviated from ideal behaviour) above eqn (iii) can't be applied.

In that case, use eqn — (i)

$$\Delta H = \Delta U + \Delta(PV)$$

* Summary -

$$\Delta H = \Delta U + \Delta(PV) \quad — (1)$$

$$\Delta H = \Delta U + \Delta n_g \times RT$$

eg. $X(s) + 2Y(g) \longrightarrow Z(g) + 3P(g)$

$$\Delta n_g = 4 - 2 = 2$$

$\Rightarrow \Delta n_g$ = Sum of st. coeff. of gaseous products — Sum of st. coeff. of gaseous reactants.

Q Calculate Heat change at constant volume when 7.8 g of liquid benzene is vaporised into gaseous benzene and at its boiling point 350 K.
Molar enthalpy of vaporisation of $C_6H_6(l)$ at 350 K is 40 KJ/mol.

Soln: $C_6H_6(l) \longrightarrow C_6H_6(g)$, $\Delta H = 40\ KJ/mol$

$$n = \frac{7.8}{78} \qquad \Delta ng = 1-0 = 1$$

$$n = 0.1\ mol$$

$$\Delta H = \Delta U + \Delta ng \times RT$$
$$40 = \Delta U + 1 \times 25/3 \times 1000 \times 360$$

$$\Delta U = 40 - 3 = 37\ KJ/mol.$$

$$(\Delta U)_{0.1\ mol} = 37 \times 0.1$$
$$= 3.7\ KJ$$

Q: 1150 Kcal of heat per mol of C_7H_{16} is released when following reaction is carried out at constant volume at 300 K.

$$C_7H_{16}(l) + 11O_2(g) \longrightarrow 7CO_2(g) + 8H_2O(l).$$

find Heat change at constant temp. & pressure.

Soln: $\Delta U = -1150\ Kcal/mol$

$$\Delta H = \Delta U + \Delta ng \times RT$$
$$\Delta H = -1150 + (-4) \times \frac{2}{1000} \times 300$$

$$\Delta H = -1152.4\ Kcal/mol$$

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Q For the reaction, $2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$, $\Delta H = -560$ kJ In one lit. vessel at 500 K, the initial pressure is 70 atm and after the reaction it becomes 40 atm at constant volume of 1 lit. Calculate change in internal energy. All the above gases show significant deviation from ideal behaviour. [Take 1 atm = 0.1 kJ]

Soln:

$$\Delta H = \Delta U + (P_2V_2 - P_1V_1)$$
$$-560 = \Delta U + 1(40 - 70) \times 0.1$$

$$\Rightarrow -560 = \Delta U - 3$$

$$\boxed{\Delta U = -557 \text{ kJ}}$$

(*) Relation between general Molar heat capacity (C) and C_v for an ideal gas :-

An ideal gas undergoes at process described by the equation PV^x = constant $\Rightarrow$ Molar heat capacity of this process is C.

From 1st law of thermodynamics,

$$dU = q + w$$
$$dU = q - PdV$$
$$\frac{C_v dT}{dT} = \frac{C dT}{dT} - \frac{P dV}{dT}$$

$\Rightarrow$

$$C = C_v + P\frac{dV}{dT} \quad \text{---} (1)$$

$$PV^x = \text{constant}$$

$$\frac{nRT}{V} \times V^x = \text{constant}$$

$$\Rightarrow \quad TV^{x-1} = \text{constant}$$

$$T \times (x-1) V^{x-2}\, dV + V^{x-1} \times dT = 0$$

$$\Rightarrow \quad \frac{dV}{dT} = \frac{-V}{T(x-1)}$$

Putting this value of $\frac{dV}{dT}$ in eq (1)

$$C = C_V + P \times \frac{dV}{dT}$$

$$C = C_V - \frac{PV}{T(x-1)}$$

For 1 mol, $PV = RT$

$$\frac{PV}{T} = R$$

$$\boxed{C = C_V - \frac{R}{(x-1)}}$$

$\{C_p - C_v = R\}$ for an ideal gas.

Q. An ideal diatomic gas undergoes a process in which ratio of pressure and volume at any instant remains constant. Find molar heat capacity for this process.

Soln: $C_V = \frac{5}{2} R.$

$P/V = \text{constant}.$

$PV^{-1} = \text{constant}$

$x = -1$

$$C = C_V - \frac{R}{x-1} = \frac{5R}{2} - \frac{R}{-2}$$

$$C = 2.5R + 0.5R = 3R \quad \text{Ans.}$$

various processes:-

(*) Calculation of ΔH, ΔU, Q, W in various processes :-

1. Isothermal process :

$T \rightarrow$ constant throughout

$dT = 0$, $\Delta T = 0$

For an ideal gas, $H = f(T)$, $U = f(T)$ $\Rightarrow \Delta H = 0$, $\Delta U = 0$

$\Delta U = Q + W$

$\Rightarrow \boxed{Q = -W}$

(a) Reversible isothermal process : $T \rightarrow$ constant

$$W_{rev.} = -\int_1^2 P_{ext}\, dV$$

$$= -\int_1^2 (P \pm dP)\, dV = -\int_1^2 P\, dV \mp \int_1^2 dP\, dV$$

(ignored)

$$\left\{ W_{rev.} = -\int_1^2 P\, dV \right\}$$

for an ideal gas, isothermal process, $PV = nRT$ constant.

$$= -\int_1^2 \frac{nRT}{V}\, dV$$

$$\left.\begin{aligned} W_{reversible} &= -nRT \ln \frac{V_2}{V_1} \\ &= -nRT \ln \frac{P_1}{P_2} \end{aligned}\right\}$$

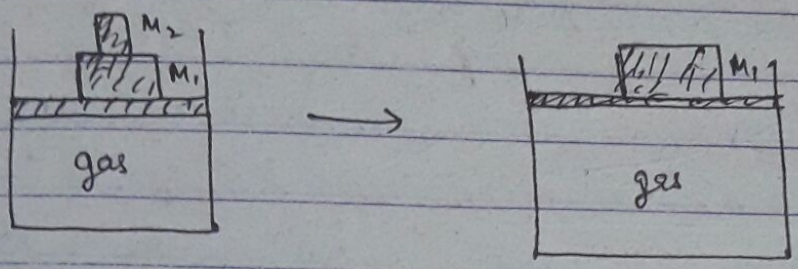
$$= -2.303\, nRT \log_{10} \frac{V_2}{V_1}$$

(b) Irreversible Isothermal process : $T \rightarrow$ constant

$$W_{irr.} = -\int_1^2 P_{ext} \times dV$$

$$= -P_{ext} \times \Delta V \quad (\text{when } P_{ext} \text{ is constant})$$

eg.



initially at eqm. → finally at eqm.

$P_{ext} = P_{atm} + \frac{(M_1+M_2)g}{A}$ Irreversible Expansion.

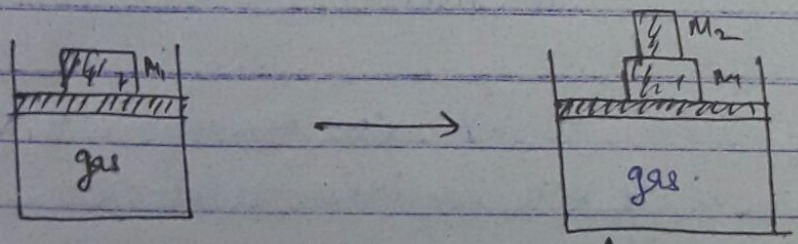
$P_{ext} = P_{atm} + \frac{M_1 g}{A}$

$$W_{irr} = -P_{ext} \times \Delta V$$

$$= -\left(P_{atm} + \frac{M_1 g}{A}\right) \times \Delta V$$

in these both cases, during the entire process, value of Pext remains constant

eg.



At eqm. → finally at eqm

Irreversible compression.

$P_{ext} = P_{atm} + \frac{(M_1+M_2)g}{A}$

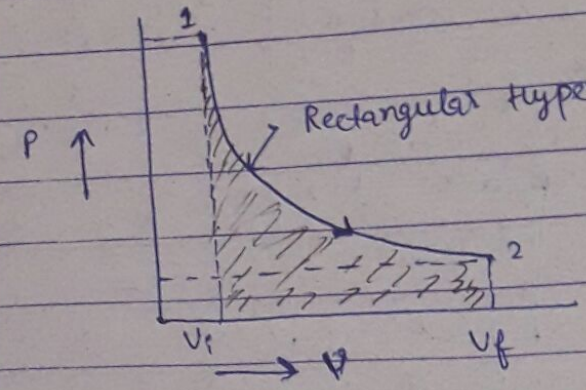
$P_{ext} = P_{atm} + \frac{M_1 g}{A}$

$$W_{irr.} = -P_{ext} \times \Delta V$$

$$= -\left(P_{atm} + \frac{(M_1+M_2)g}{A}\right) \times \Delta V$$

* Graphical representation :

(a) Reversible isothermal process : $PV = \text{constant} = nRT$ [for an ideal gas]

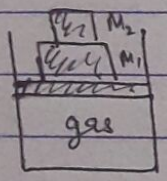


$$W_{rev.} = -\int_1^2 P\,dV$$

Area under P v/s V curve represents magnitude of work done.

we know exact equation of curve
so we can plot a smooth curve

(b) Irreversible isothermal process :-

eg. 

→ In irreversible process, eqⁿ between system & surroundings exist only at initial and final states and not during the process.

Imp.

→ No relation exists between P & V or P & T or V & T in irr. process.
So, smooth curve (exact curve) can't be plotted.

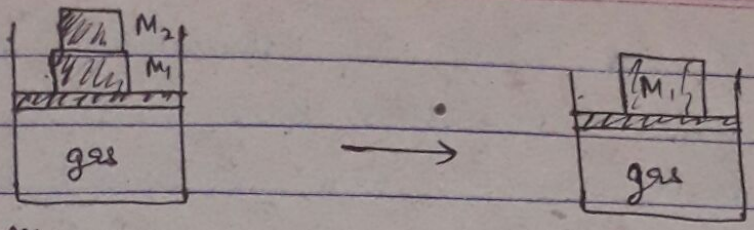
hence, whenever we are given an eqn. of the process it auto implies reversible process

* If expansion or compression is against (or by) a constant external pressure, then it is irreversible process.

→ We can show work done in irr. process by P v/s V graph (approximate graph)

it does not represent the process
it is simply used to account the work done

eg.



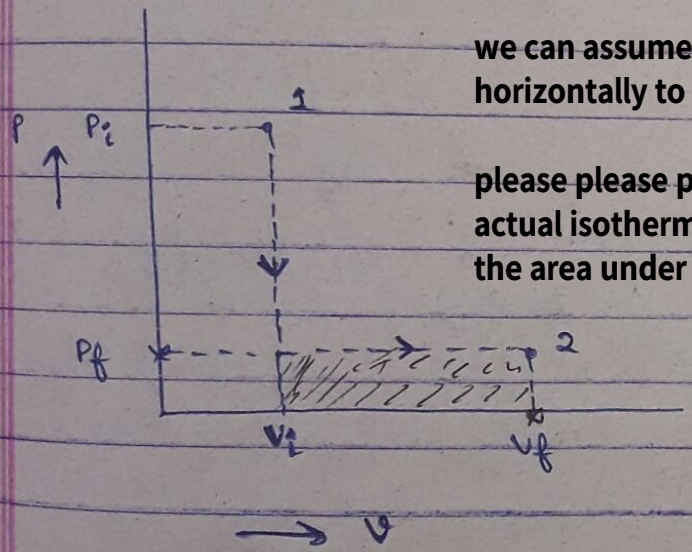
at eqm. initially

finally at eqm.

Irreversible Expansion.

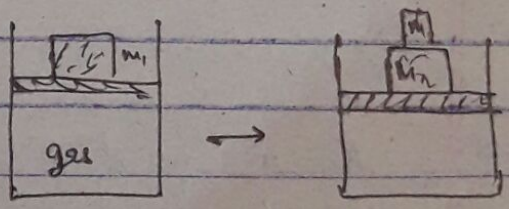
$$W_{irr.} = -P_{ext} \times \Delta V$$

$$= -\left(P_{atm} \times \frac{M_1 g}{A}\right) \times \Delta V$$



we can assume we came down vertically from 1 and then went horizontally to 2

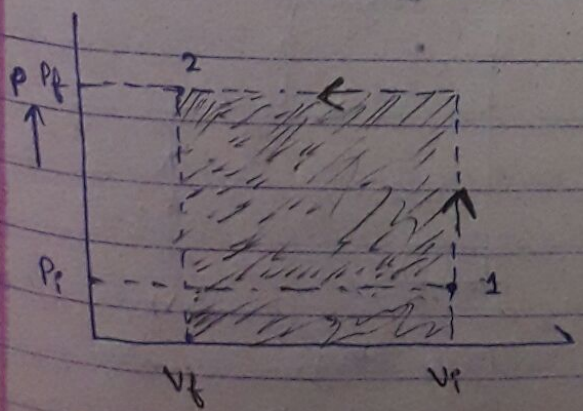
please please please remember that this does not represent the actual isothermal process, purpose of this graph is to just find out the area under the curve, that represents the work done



eg. Irreversible Compression :

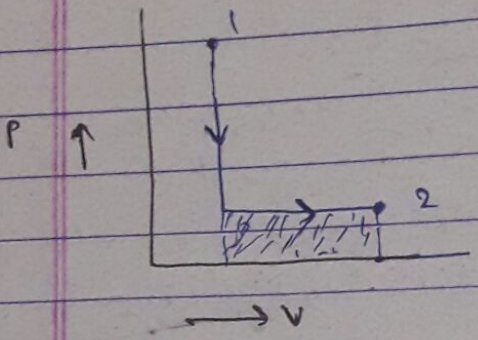
$$W_{irr} = -P_{ext} \times \Delta V$$

$$= -\left(P_{atm} + \frac{(M_1+M_2)g}{A}\right) \times \Delta V$$

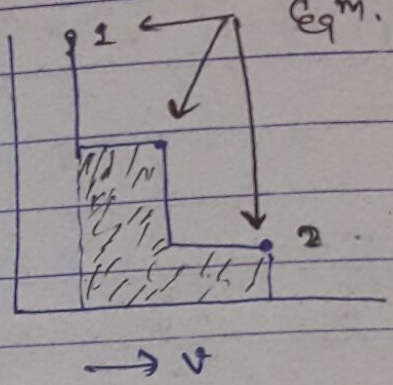


vertically up and then moving left

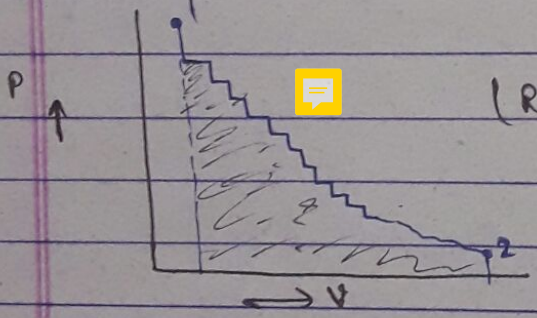
Note: (1) As the number of expansion steps increase, magnitude of work done increases and becomes maximum for ∞ stage expansion (rev. expansion)



Single stage expansion

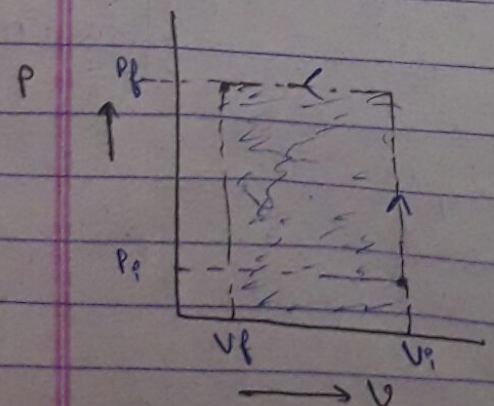


Two stage expansion.

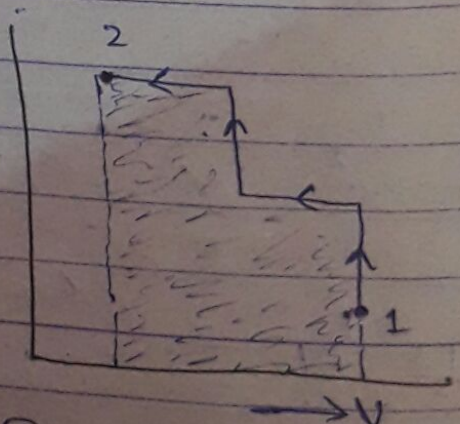


(Rev. expansion ∞ stage)

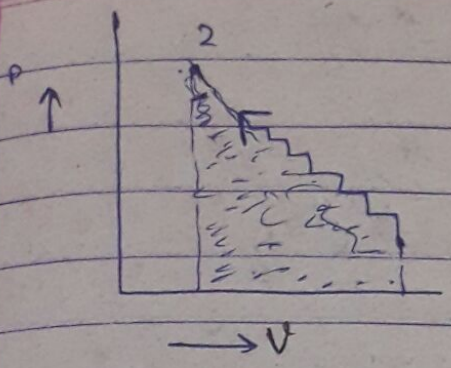
→ As the number of compression steps increase, work done decreases and reaches minimum for ∞ stage compression.



Single stage compression



Two stage Compression.



∞ stage (or Reversible compression)

Imp Note: Between two states,

$W_{irr.} > W_{rev.}$ [For both expansion & compression.]

applicable for isothermal as well as adiabatic

Q. 2 mol of an ideal gas is isothermally expanded from 2 atm to 1 atm at 300K. Calculate Q, W, ΔU, ΔH if

a. Process is reversible

b. Process is irreversible & expansion occured against a constant external pressure of 1 atm.

Soln: For an ideal gas, $U = f(T)$, $H = f(T)$ } $\Delta U = 0$, $H = 0$

$\Delta U = Q + W \Rightarrow Q = -W$

a) Rev. iso. process: $W_{rev} = -nRT \ln \frac{P_1}{P_2}$

$= -2 \times R \times 300 \ln 2/1$

$= -600R \times 0.7$

$= -420R$

$Q_{rev} = -W_{rev.} = +420R$

ln2 = 0.69
= 0.7 (approx.)

(b) Irr. iso. process: $W_{irr} = -\int_1^2 P_{ext} \times dV$

$= -P_{ext} \times \Delta V$

$= -1\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right)$

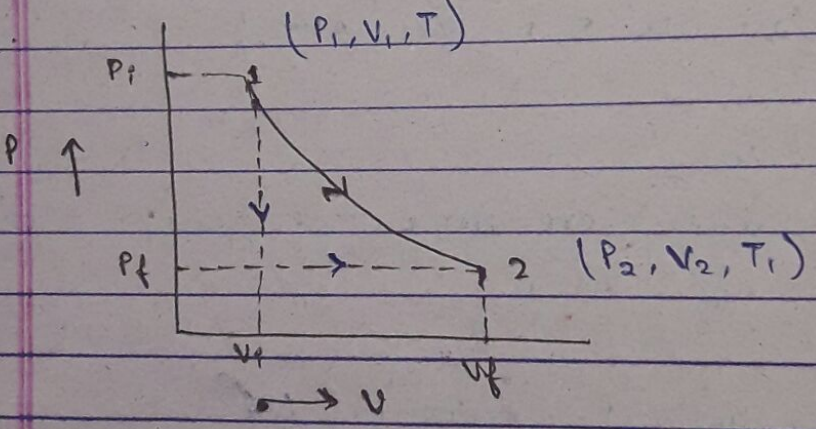
$= -2 \times R \times 300\left(\frac{1}{1} - \frac{1}{2}\right)$

$= -600R \times 0.5$

$= -300R$

Here we observe $|W_{rev}|_{iso} > |W_{irr.}|_{iso}$

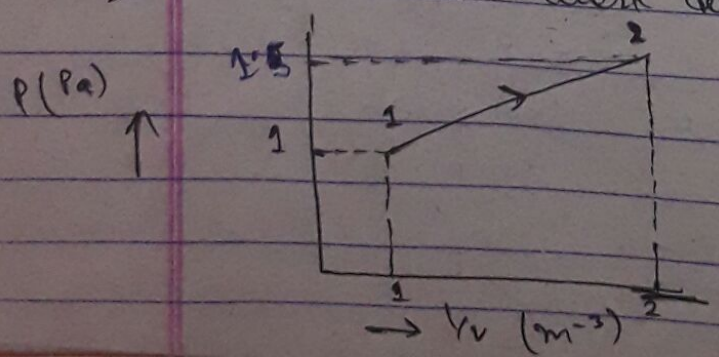
Gmp



$PV = nRT$

** Both reversible and irreversible isothermal processes end up at same final state if expanded or compressed to same final volume.

Q Calculate the work done if an ideal gas undergoes this reversible process.



Soln :

$P_1V_1 = 1 \times 1$

$P_2V_2 = 0.5 \times 0.5 = 0.75$

$\Rightarrow P_1V_1 \neq P_2V_2$ (Not an isothermal process)

$P = m \times \frac{1}{V} + c$

$1 = m \times 1 + c$

$1.5 = m \times 2 + c$

$m = 0.5 \,, \; c = 0.5$

$\Rightarrow P = \frac{1}{2V} + \frac{1}{2}$

$W = -\int_1^2 P\,dV$

$= -\int_1^{0.5} \left(\frac{1}{2V} + \frac{1}{2}\right) dV$

$W = -\frac{1}{2} \ln \frac{0.5}{1} - \frac{1}{2}(0.5 - 1)$

$W = \frac{1}{2} \ln 2 + \frac{1}{4}$

$W = \frac{0.7}{2} + 0.25$

$W = 0.35 + 0.25$

$W = 0.6$ Joule Ans.

② Isochoric Process :-

V → constant throughout

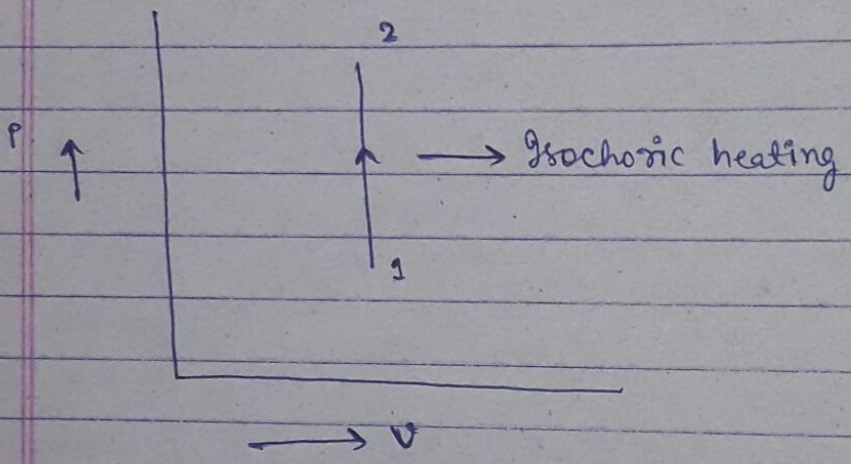
$dV = 0$, $\Delta V = 0$

$$W = -\int_1^2 P\,dV = 0$$

for an ideal gas, $Q_V = \Delta U = nC_v(T_2 - T_1)$

$W = 0$

$\Delta H = nC_p(T_2 - T_1)$



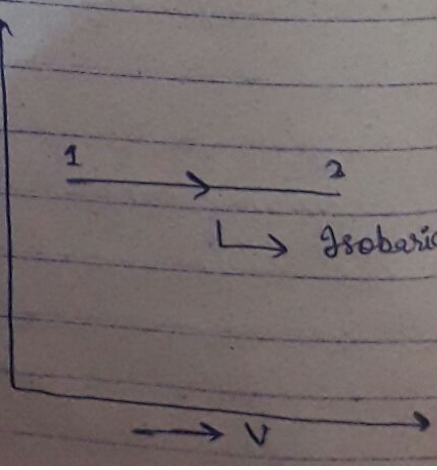
③ Isobaric process :

P → constant throughout

$dP = 0$, $\Delta P = 0$

$$W = -\int_1^2 P_{ext}\,dV = -P(V_2 - V_1)$$

for an ideal gas
$$\begin{cases} Q_p = \Delta H = nC_p(T_2 - T_1) \\ \Delta U = nC_v(T_2 - T_1) \end{cases}$$



(4) Adiabatic process :

No heat exchange between system & surroundings.
$(q = 0)$

for an ideal gas, $\Delta H = nC_p(T_2 - T_1)$

$\Delta U = nC_v(T_2 - T_1)$

$\Delta U = \cancel{q}^{0} + w$

$w = \Delta U = nC_v(T_2 - T_1)$

If T_1 and T_2 are known in adiabatic process, apply these formulae.

→ If either of initial or final temperature is not known, then our first task would be to find unknown temperature.

(i) Reversible adiabatic process : (For an IG)

$q_{rev} = 0$, $dU = \cancel{q_{rev.}}^{0} + w_{rev.}$

$nC_v dT = -PdV$

$\left\{ \begin{array}{l} C_p - C_v = R \\ \gamma = \frac{C_p}{C_v} \end{array} \right.$

$n \times \frac{R}{\gamma - 1} \times dT = -\frac{nRT}{V} dV$

$$\int_{T_1}^{T_2} \frac{dT}{T} = -(\gamma - 1) \int_{V_1}^{V_2} \frac{dV}{V}$$

$$\Rightarrow \ln \frac{T_2}{T_1} = -(\gamma - 1) \ln \left(\frac{V_2}{V_1} \right)$$

$$\Rightarrow \ln \frac{T_2}{T_1} = \ln \left(\frac{V_1}{V_2} \right)^{\gamma - 1}$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \Rightarrow T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1} = \text{Constant}$$

$$\Rightarrow \boxed{TV^{\gamma-1} = \text{constant}}$$

$$\frac{PV}{nR} \times V^{\gamma-1} = \text{constant}$$

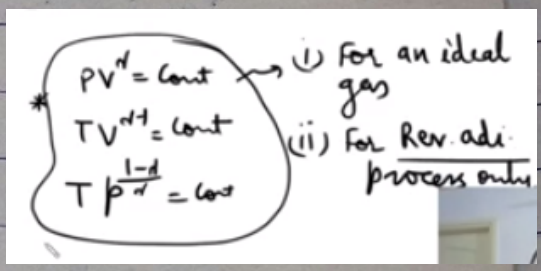
applicable

(i) for an ideal gas
(ii) for Rev. adiabatic process only

$$\Rightarrow \boxed{PV^{\gamma} = \text{constant}}$$

$$P \times \left(\frac{nRT}{P}\right)^{\gamma} = \text{constant}$$

$$\Rightarrow \boxed{TP^{(1-\gamma)/\gamma} = \text{constant}}$$



→ NOT APPLICABLE FOR IRREVERSIBLE ADIABATIC PROCESS

(ii) Irreversible adiabatic process :-

$Q_{irr.} = 0$, $\Delta U = \cancel{Q_{irr}}^{0} + W_{irr.}$

$$\boxed{nC_v(T_2 - T_1) = -P_{ext}(V_2 - V_1)}$$

OR

$$\boxed{nC_v(T_2 - T_1) = -P_{ext}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)}$$

- for an ideal gas
- only for irr. adiabatic process.

Q. 2 mol of an ideal triatomic non-linear ideal gas is adiabatically expanded from 16 atm & 300K to a final pressure of 1 atm.

Calculate Q, W, ΔU, ΔH, if

a. Process is rev.

b. Process is irreversible & expansion occured against a constant external pressure of 1 atm.

Soln:

16 atm $\longrightarrow$ 1 atm, $T_1 = 300\,K$

(a.) Rev. adi. process, $TP^{\frac{1-\gamma}{\gamma}} = \text{constant}$

$$\frac{T_2}{T_1} = \left(\frac{P_1}{P_2}\right)^{\frac{1-\gamma}{\gamma}}$$

Triatomic non-linear ideal gas,

$C_V = 3R$, $C_P = 4R$

$\gamma = 4/3$

$$\Rightarrow \frac{T_2}{300} = \left(\frac{16}{1}\right)^{\frac{1-4/3}{4/3}}$$

$$\Rightarrow \frac{T_2}{300} = (2^4)^{-1/4} = \frac{1}{2} \Rightarrow (T_2)_{rev} = 150\,K.$$

$Q = 0$, $\Delta U = W = nC_V(T_2 - T_1)$

$= 2 \times 3R\,(150 - 300)$

$= -900R$

$\Delta H = nC_P(T_2 - T_1)$

$= 2 \times 4R\,(-150)$

$= -1200R$

(b) Irr. adiabatic process.

$$\Delta U = W_{irr} \Rightarrow nC_v(T_2 - T_1) = -P_{ext}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$$

$$\Rightarrow 3R(T_2 - 300) = -1\left(\frac{RT_2}{1} - \frac{R \times 300}{16}\right)$$

$$\Rightarrow 4T_2 = 900 + 300/16$$

$$\Rightarrow T_2 = 229.67\ K \approx 230\ K$$

$$Q = 0,\ \Delta U = W_{irr} = nC_vR(T_2 - T_1) = 2 \times 3R(230 - 300) = -420R$$

$$\Delta H = nC_p(T_2 - T_1) = 2 \times 4R(-70) = -560R$$

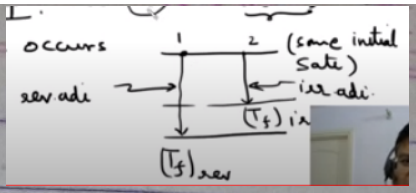
* Observation:

(1) $(T_f)_{irr,adi.} > (T_f)_{rev.adi.}$ for same final volume and final pressure

This holds true for both expansion & compression.

(2) $|W|_{rev.adi.exp^n} > |W|_{irr.adi.exp^n}$.

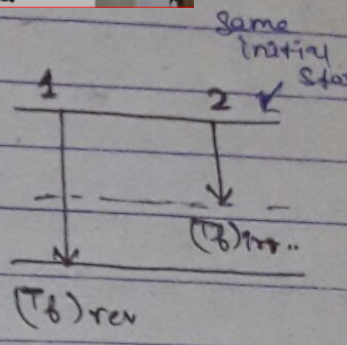
* Rev. process is more efficient than irr. process.



* Conclusion:

→ for adiabatic expansion,

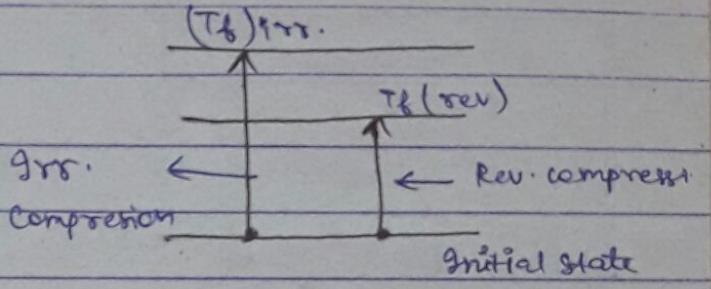
$$\Delta U = W = n C_v (T_2 - T_1)$$



* In adi. expansion cooling occurs.

→ for adiabatic compression,

$$\Delta U = W = n C_v \Delta T$$



* Heating occurs.

V. Imp

$$* \left\{ (T_f)_{irr.adi} > (T_f)_{rev.adi} \right\}$$

applicable when:

→ for both expansion and compression.

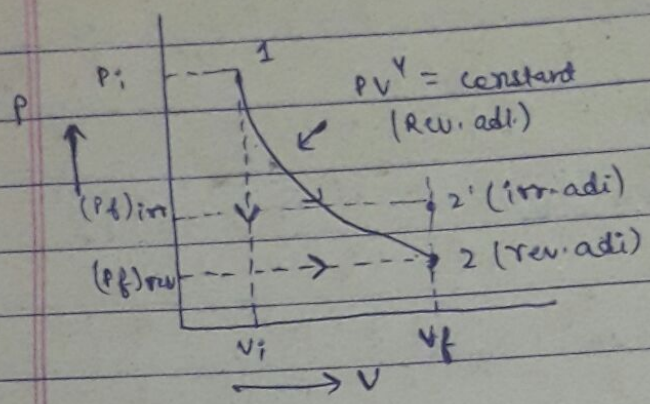
→ Both reversible and irreversible adiabatic process start from same initial state.

→ Both process end up at same final volume or same final pressure.

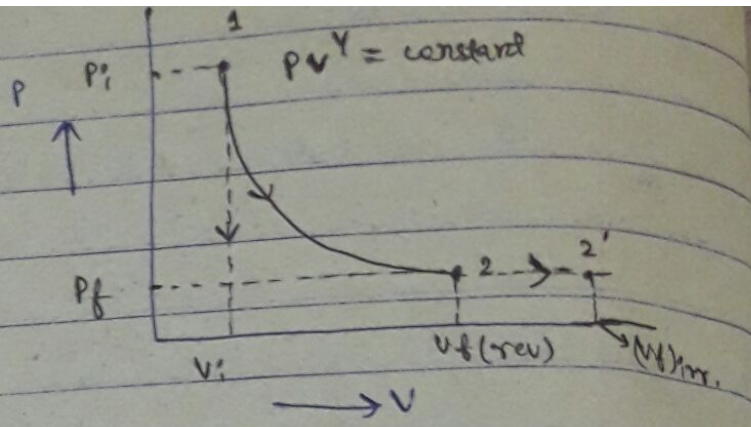
* Graphical Comparison between rev. and irr. adiabatic process:
(In P v/s V graph for ideal gas)

a. for expansion : $q = 0$, $\Delta U = W = n C_v \Delta T$

in both these curves, just use (Tf)irr. > (Tf)rev.



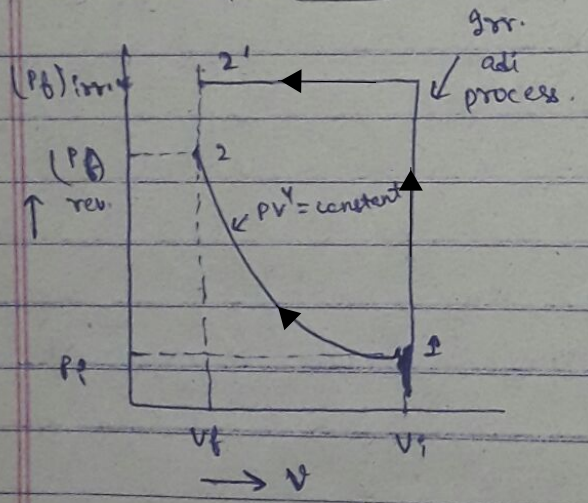
for same final volume



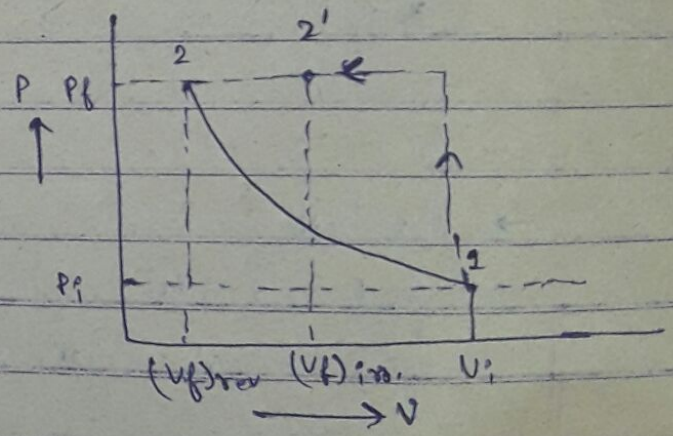
for same final pressure.

$PV = nRT$

(b) Compression:



for same final volume.



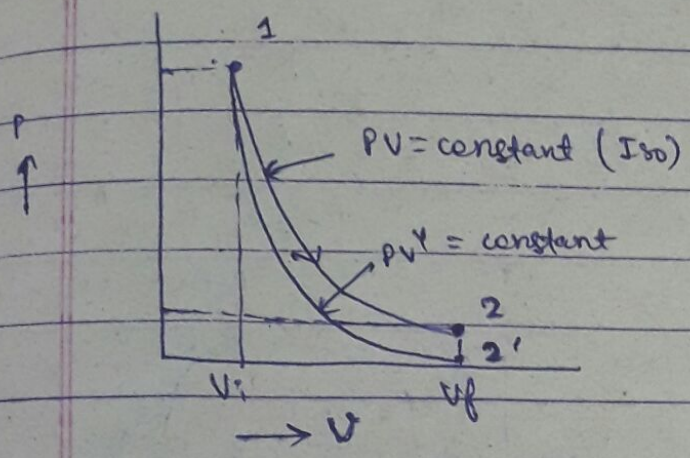
for same final pressure.

(*) Graphical comparison between reversible isothermal ($PV = k$) and reversible adiabatic (PV^γ = constant).

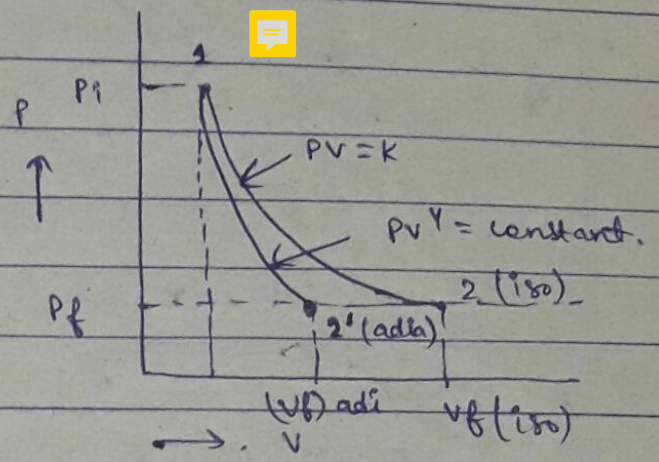
Reversible isothermal,	Reversible adiabatic
$PV = k$	PV^γ = constant
$PdV + VdP = 0$	$P \times \gamma V^{\gamma-1} dV + V^\gamma dP = 0$
$\Rightarrow \frac{dP}{dV} = \frac{-P}{V}$	$\Rightarrow \frac{dP}{dV} = -\gamma \times P/V$ $\gamma > 1$

Imp. → Slope of P v/s V graph rev. adi has greater steepness than that in rev. isothermal process.

(a) For expansion :

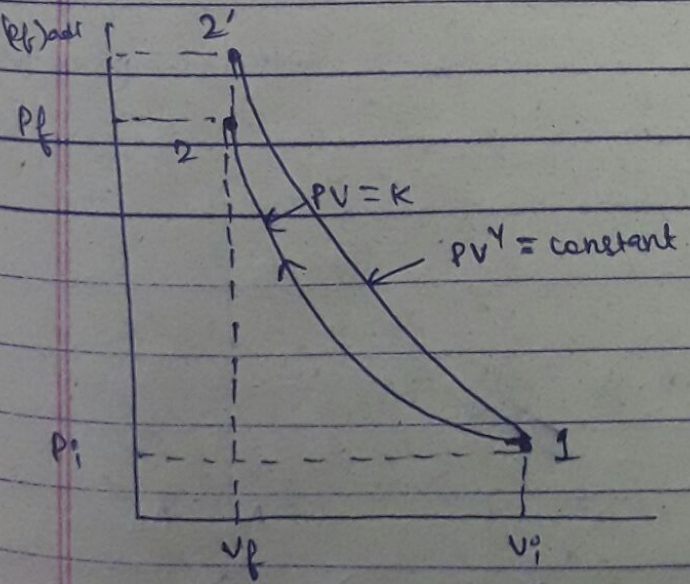


for same final volume

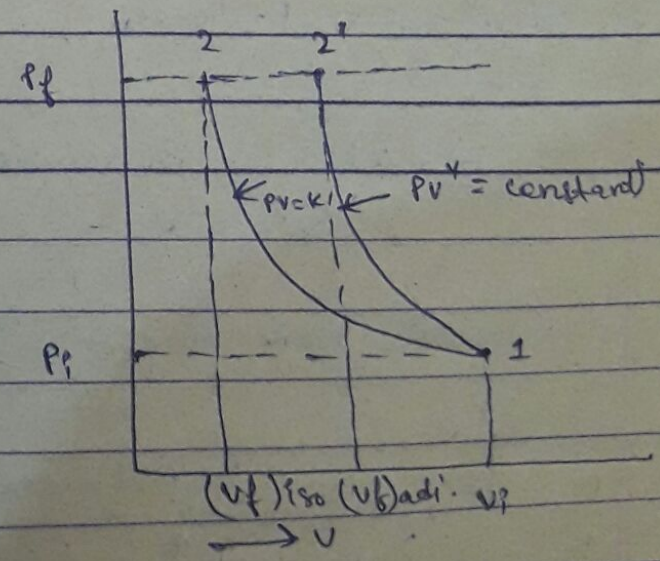


for same final pressure.

(b) For compression :



for same final volume



for same final pressure.

Cyclic Process:

(*) Cyclic process :

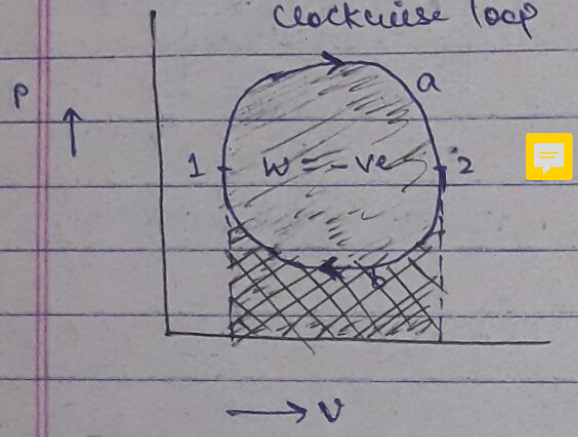
Change in state function in cyclic process = 0.

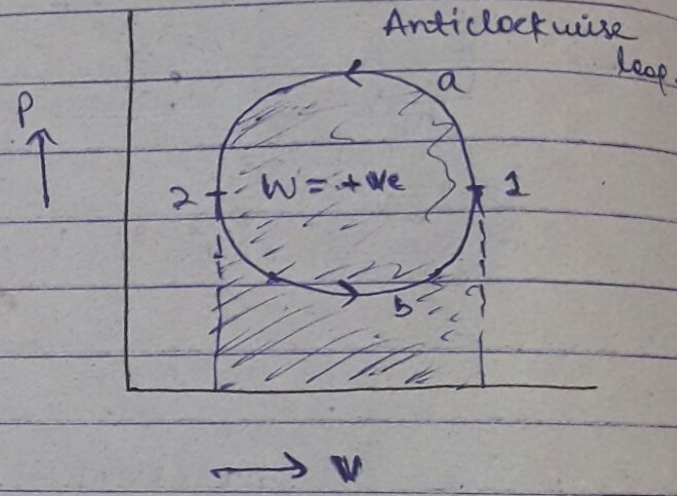
$\therefore \Delta\phi = 0$

$\Delta T = 0$, $\Delta P = 0$, $\Delta S = 0$, $\Delta H = 0$, $\Delta U = 0$ etc

as initial and final states are same.

($\Delta T = 0$, $dT \neq 0$)





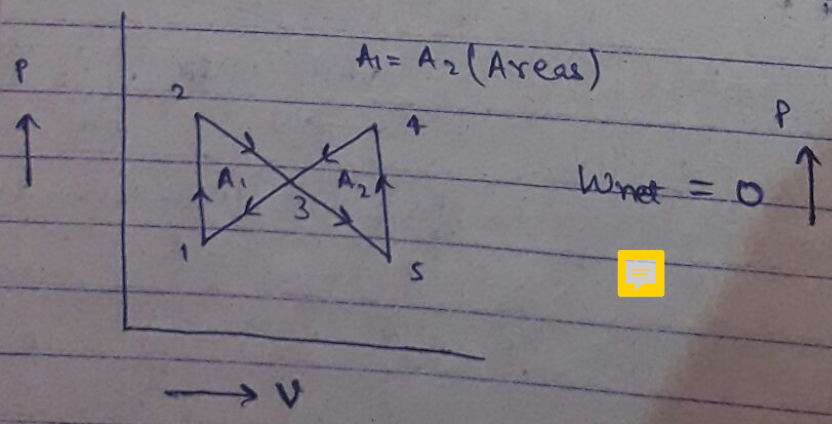
W_{net} = -ve (expansion)

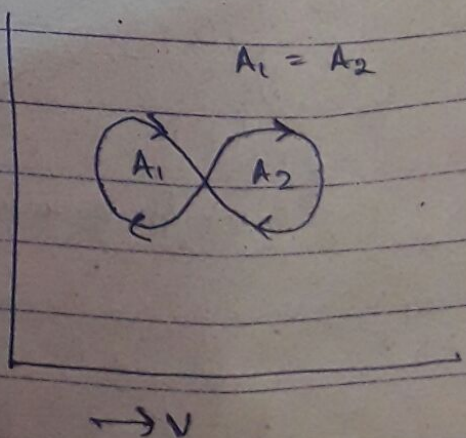
W_{net} = +ve (compression)

$\Rightarrow \Delta U = 0$ in cyclic process,

$Q = -W$

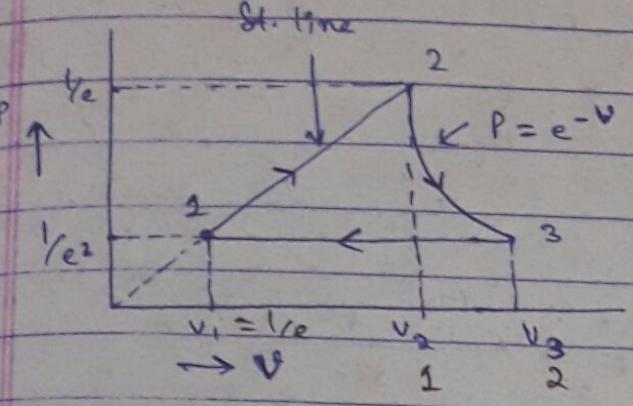
$\Rightarrow$ work done may be zero in a cyclic process.





W_{net} = -ve (expansion)

Q. A gas is subjected to the following reversible processes as shown. Find work done.



Soln :

$$P = e^{-V}$$

$$\frac{1}{e} = e^{-V_2} \Rightarrow V_2 = 1$$

$$\frac{1}{e^2} = e^{-V_3} \Rightarrow V_3 = 2$$

Path, 1-2, $P = m \times V$

$$\frac{1}{e} = m \times 1 \Rightarrow P = \frac{1}{e} \times V$$

$$\frac{1}{e^2} = \frac{1}{e} \times V_1 \Rightarrow V_1 = \frac{1}{e}$$

$$W_T = W_{(1-2)} + W_{(2-3)} + W_{(3-1)}$$

$$= -\int_1^2 P\,dV - \int_2^3 P\,dV - \int_3^1 P\,dV$$

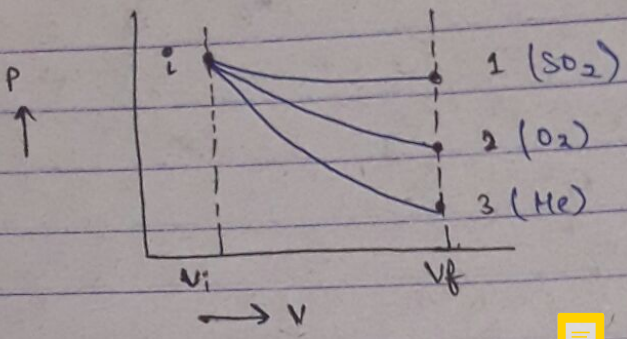
$$= -\int_{1/e}^{1} \frac{1}{e} V\,dV - \int_1^2 e^{-V}\,dV - 0 \quad \left[\because \frac{1}{e^2}\left(\frac{1}{e} - 2\right)\right]$$

$$= -\frac{1}{2e}\left[1 - \frac{1}{e^2}\right] + \left(e^{-2} - e^{-1}\right) - \frac{1}{e^3} + \frac{2}{e^2}$$

$$= -\frac{1}{2e} + \frac{1}{2e^3} + \frac{1}{e^2} - \frac{1}{e} - \frac{1}{e^3} + \frac{2}{e^2}$$

Questions :

Q. Three gases (O_2, He, SO_2) undergo the reversible adiabatic expansion from same initial point upto same final volume.



Match end points 1, 2, 3 with given gases.

PV^x = constant.

greater the value of x, steeper is the curve

Soln : O_2, He, SO_2

⇒ γ is max. for monoatomic gas.

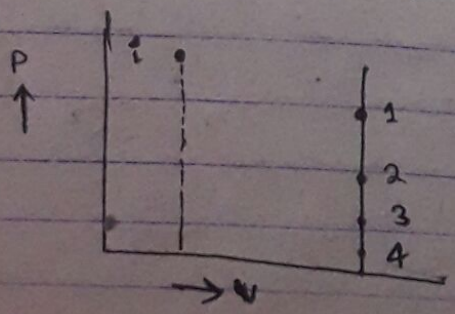
$He > O_2 > SO_2$

decreasing γ →

1 → SO_2 , 2 → O_2 , 3 → He

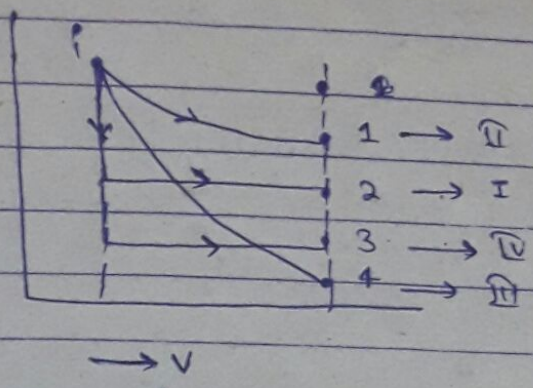
Ans Q. If 4 identical samples of an ideal gas initially at similar state (P_0, V_0, T_0) are allowed to expand to double their volume by four different process.

1. Isothermal irreversible process.
2. Rev. process, P^2V = constant.
3. Rev. ad. process
4. Irr. ad. expansion, against constant external pressure.



Match processes I, II, III & IV with end points 1, 2, 3, 4

Soln :



$PV^{\gamma} = \text{const}$ | $P^{\gamma}V = \text{const}$

$PV^{\gamma/2} = \text{const}$

Q. A certain amount of a gas is expanded from (1L, 10 atm) to (4L, 5 atm) against a constant external pressure of 1 atm. If initial temp. of gas is 300 K and heat capacity of process is 50 J/°C.
find ΔH during the process. [Take 1 atm = 10^5 Pa]

Soln:

1L, 10 atm, 300K ⟶ 4L, 5 atm, $T_2 = 600K$.

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow \frac{10 \times 1}{300} = \frac{5 \times 4}{T_2}$$

$$T_2 = 600k$$

$$Q = C \times \Delta T = 50 \times (300)$$
$$= 15 KJ.$$

$$W = -P_{ext} \times \Delta V = -1(4-1) = -3 \text{ lit atm}$$
$$= -3 \times 10^{-3} m^3 \times 1 \times 10^5 Pa$$
$$= -300 J$$

$$\Delta U = Q + W$$
$$= 15000 - 300 = 14700 J$$

$$\Delta H = \Delta U + \Delta(PV)$$
$$= 14700 + (5 \times 4 - 1 \times 10) \times 10^{-3} \times 10^5 = 15700 \text{ Joule}$$

(*) Drawbacks of 1st law of Thermodynamics:-

1. No information about the feasibility of a process or reaction.
2. No information to what extent a reaction proceed.

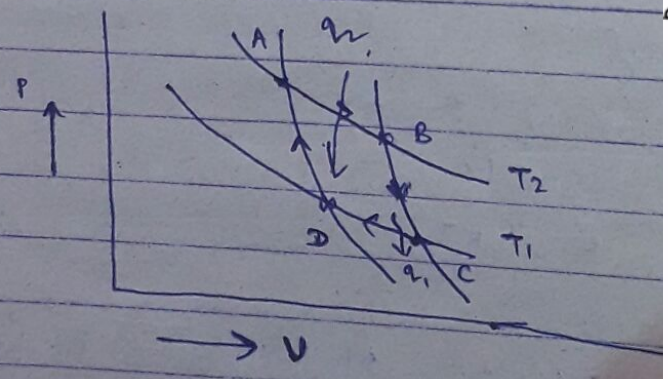
→ Above are well answered by 2nd law of T.D.

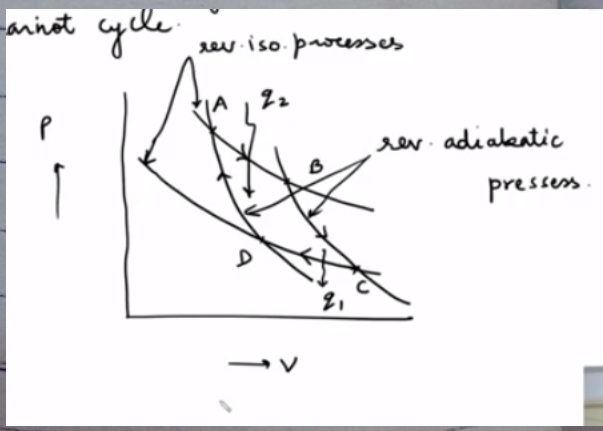
* Introduction of entropy by carnot cycle :-

→ Carnot cycle is an idealized cyclic process consisting of two reversible isothermal & two reversible adiabatic processes.

→ The cyclic processes are the bases for working of power plants (eg. Thermal power plant) to convert heat into work energy. heat is the unorganised form of energy, work is the organised form or desired form of energy

→ Consider ideal gas to be the working fluid in carnot cycle.





Carnot cycle

$$\eta_{Carnot} = \frac{\text{Work delivered by heat engine}}{\text{Heat input into heat engine.}}$$

AB process: Reversible isothermal

$\Delta U = 0, \quad q_{AB} = -W_{AB}$

$= nRT_2 \ln \frac{V_B}{V_A}$

BC process: Rev. Adiabatic

$q_{BC} = 0, \quad W_{BC} = \Delta U$

$= nC_V (T_1 - T_2)$

CD process: Rev. isothermal

$\Delta U = 0, \quad q_{CD} = -W_{CD}$

$= nRT_1 \ln \frac{V_D}{V_C}$

DA process: Rev. adiabatic

$q_{DA} = 0, \quad W_{DA} = \Delta U$

$= nC_V (T_2 - T_1)$

$W_T = W_{AB} + W_{BC} + W_{CD} + W_{DA}$

$= -nRT_2 \ln \frac{V_B}{V_A} + nC_V(T_1 - T_2) + (-nRT_1 \ln \frac{V_D}{V_C}) + nC_V(T_2 - T_1)$

$\Rightarrow$ Process BC:- $T_2 V_B^{\gamma-1} = T_1 V_C^{\gamma-1}$

adiabatic reversible

$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_C}{V_B}\right)^{\gamma-1}$

$\Rightarrow$ Process DA: $T_1 V_D^{\gamma-1} = T_2 V_A^{\gamma-1}$

$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_D}{V_A}\right)^{\gamma-1}$

$\Rightarrow \frac{V_C}{V_B} = \frac{V_D}{V_A} \Rightarrow \frac{V_B}{V_A} = \frac{V_C}{V_D}$

$$W_T = -nRT_2 \ln \frac{V_B}{V_A} + \left(-nRT_1 \ln \frac{V_D}{V_C}\right)$$

$$= -nRT_2 \frac{V_B}{V_A} + nRT_1 \ln \frac{V_B}{V_A}$$

A

$$\boxed{W_T = -\left\{nR \ln \frac{V_B}{V_A}\right\}\left(T_2 - T_1\right)}$$

$$\eta_{carnot} = \frac{-W_T}{q_{AB}} = \frac{+ \cancel{nR} \ln \frac{V_B}{V_A} \times (T_2 - T_1)}{\cancel{nRT_2} \ln \frac{V_B}{V_A}}$$

$$\left\{ \eta_{carnot} = \left(\frac{T_2 - T_1}{T_2}\right) \right\}$$

Carnot cycle means all are reversible process.

-ve taken to make the term +ve as WD = -ve

$$\eta_{carnot} = \frac{-W_T}{q_2}$$

$$\Rightarrow \boxed{\eta_{carnot} = \frac{q_2 + q_1}{q_2} = \frac{T_2 - T_1}{T_2}}$$

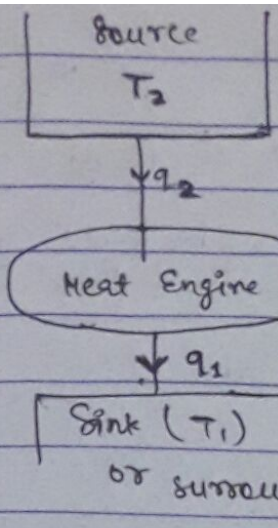
In a cyclic process

$\Delta U = 0$

$q_{total} + W_{total} = 0$

$q_{total} = -W_{total}$

$(q_2 + q_1) = -W_{total}$



thermal power plant, turbine, boiler etc.

$T_2 > T_1$

$\eta_{carnot} = \frac{T_2 - T_1}{T_2}$

Source : High temperature reservoir. (Burning of Coal)

Sink : low temperature reservoir. (Surrounding)

$$\eta_{carnot} = \frac{q_2(rev) + (q_1)_{rev}}{q_2(rev)} = \frac{T_2 - T_1}{T_2}$$

$$1 + \frac{(q_1)_{rev}}{(q_2)_{rev}} = 1 - \frac{T_1}{T_2}$$

$$\frac{(q_1)_{rev}}{T_1} + \frac{(q_2)_{rev}}{T_2} = 0$$

$$\frac{(q_1)_{rev}}{T_1} + 0 + \frac{(q_2)_{rev}}{T_2} + 0 = 0$$

$\frac{(q_1)_{rev}}{T_1}$ for CD ; $+0$ for BC ; $\frac{(q_2)_{rev}}{T_2}$ for AB ; $+0$ for DA

$$\sum \frac{q_{rev}}{T} = 0$$

$$\oint q_{rev}/T = 0$$

Since cyclic integration of state function is zero.

So $\frac{q_{rev}}{T} = ds$

where s is called Entropy of System

unit : J/K or cal/K

$$ds = q_{rev}/T$$

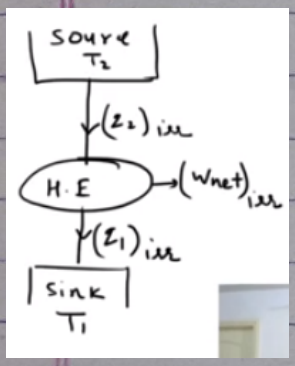
$$\Delta S = \int_1^2 \frac{q_{rev}}{T}$$

$$ds = \frac{(q_{rev}}{T}$$

→ If one of the process or more processes in cyclic process is irreversible.

$$\eta_{irr.\ process} < \eta_{rev.\ process}$$

$$\eta_{irr.\ cyclic\ process} < \eta_{carnot}$$



$$\frac{(q_2)_{irr.} + (q_1)_{irr.}}{(q_2)_{irr.}} < \frac{T_2 - T_1}{T_2}$$

$$1 + \frac{(q_1)_{irr}}{(q_2)_{irr}} < 1 - \frac{T_1}{T_2}$$

$$\frac{(q_1)_{irr}}{T_1} + \frac{(q_2)_{irr}}{T_2} < 0$$

$$\left\{\frac{(q_1)_{irr}}{T_1} + 0 + \frac{(q_2)_{irr}}{T_2} + 0\right\} < 0$$

$$\oint \frac{(q)_{irr}}{T} < 0$$

So, $\frac{q_{irr}}{T} \neq ds$

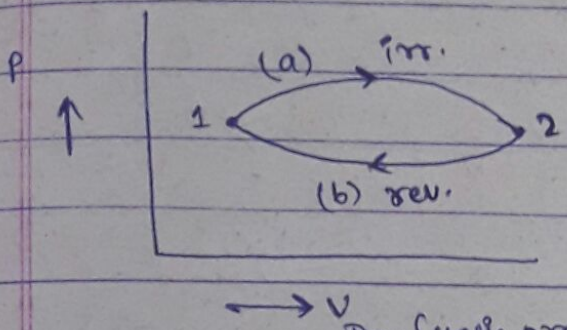
gmp ** Entropy is a state function & change in entropy is a definite quantity between two states. and does not depend on path.

→ Change in entropy ~~does~~ is a definite quantity between two states irrespective of rev. or irr. process & is only equal to $\int_1^2 q/T$ when process is reversible.

$$\oint \frac{q}{T} \leq 0$$

'<' sign for irr. cyclic process
'=' sign for rev. cyclic process.

* Clausius inequality :-



Irr. Cyclic process

$$\oint \frac{q_{irr}}{T} < 0$$

$$\int_1^2 \frac{q_{irr}}{T} + \int_2^1 \frac{q_{rev}}{T} < 0$$

$$\Rightarrow \int_1^2 \frac{q_{rev\,irr.}}{T} - \int_1^2 \frac{q_{rev}}{T} < 0$$

$$\Rightarrow \int_1^2 \frac{q_{rev\,irr.}}{T} < \int_1^2 ds$$

$$\Rightarrow \frac{q_{irr}}{T} < ds$$

$$\boxed{\frac{q}{T} \leq ds} \quad \text{OR} \quad \boxed{\frac{dq}{T} \leq ds}$$

Clausius inequality

'<' for irr.

'=' for rev.

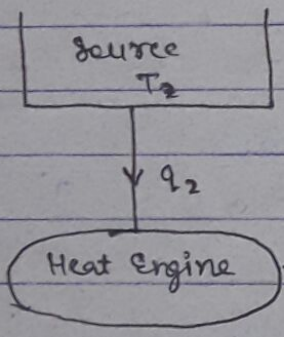
(*) 2nd law of Thermodynamics :-

Statement - I (Kelvin Planck Statement)

It is impossible to construct a heat engine working in a cyclic process whose sole effect is to convert heat into work without the simultaneous transfer of heat from high temp. body (source) to a low temp. body (sink)

OR

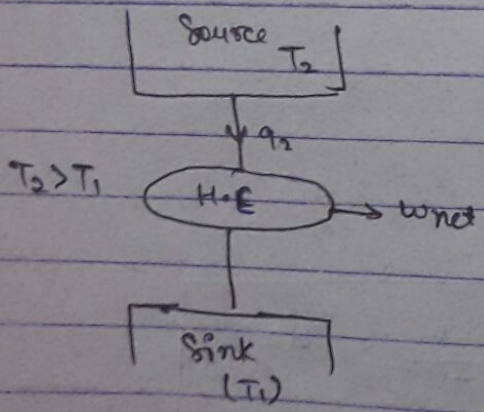
Heat can't be completely converted into work.



Impossible arrangement

$q_2 = |W|$

$\eta_{H.E} = 100\%$ (impossible)



Actual Arrangement of working of a power plant.

→ Heat : Unorganised or Random form of energy

Work : Organised or Ordered form of energy.

Statement -II

Entropy of universe remains constant in a reversible process, but it increases in an irreversible or spontaneous process.

* Spontaneous process :
which happens on its own without any external help. Spontaneous may be very fast (like neutralization rxn.) or very slow (eg. rusting of iron in moist atmosphere air)

OR

" Entropy of universe is continuously increasing "

If $(\Delta S_{sys.} + \Delta S_{surr.}) > 0$, spontaneous or irr. process

If $(\Delta S_{sys.} + \Delta S_{surr.}) = 0$, Reversible process
[Eqm. state)

If $(\Delta S_{sys.} + \Delta S_{surr.}) < 0$, process is non-occuring or non-spontaneous.

criterion of spontaneity

* Determination of ΔS surrounding :

→ Since surrounding is much larger than the size of system, P and T of surroundings is considered constant due to some process occuring in a system.

→ Process w.r.t surroundings is considered reversible even if the process occuring in system is irreversible.

$$\Delta S_{surr.} = \int_1^2 \frac{(q_{surr.})_{rev}}{T_{surr.}} = \frac{Q_{surr.}}{T_{surr.}}$$

$$\boxed{\Delta S_{surr.} = \frac{-Q_{sys.}}{T_{surr.}}}$$

$$Q_{surr} = -Q_{sys}$$

⊛ Calculation of entropy change ($\Delta S_{sys.}$) for system :-

(a) For an ideal gas undergoing any process :

$$\Delta S_{sys.} = \int_1^2 \frac{q_{rev}}{T} = \int_1^2 \frac{dU - W_{rev.}}{T}$$

$$= \int_1^2 \frac{nC_v dT - (-PdV)}{T}$$

$$= n\int_1^2 \frac{C_v dT}{T} + \int_1^2 \frac{nR}{V} dV$$

$$\boxed{\Delta S_{sys.} = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}} \quad \text{—— (1)}$$

→ for an ideal gas,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\Rightarrow \frac{V_2}{V_1} = \frac{P_1}{P_2} \times \frac{T_2}{T_1}$$

Now, $\Delta S_{sys.} = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2} + nR \ln \frac{T_2}{T_1}$

$$\boxed{\Delta S_{sys.} = nC_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}}$$

↳ applicable for both rev. & irr. processes.

→ Isothermal process, $dT = 0$, $\Delta T = 0$

$$\Delta S_{sys.} = nR \ln \frac{V_2}{V_1} = nR \ln \frac{P_1}{P_2}$$

→ Isochoric process, $dV = 0$

$$\Delta S_{sys.} = nC_v \ln \frac{T_2}{T_1}$$

→ Isobaric process, $dp = 0$

$$\Delta S_{sys.} = nC_p \ln \frac{T_2}{T_1}$$

→ for adiabatic process :

(i) Reversible adi process, $q_{rev} = 0$

$$(ds)_{sys} = \frac{q_{rev}}{T} = 0$$

$$q_{surr} = 0, \quad (ds)_{surr.} = \frac{q_{surr.}}{T} = 0$$

$$\Delta S_T = 0$$

called Isentropic process.

(ii) Irreversible adiabatic process :

$$Q_{irr.} = 0$$

$$q_{surr} = 0$$

$$\Delta S_{surr.} = \int_1^2 \frac{q_{surr}}{T} = 0$$

From 2nd law of T.D, for an irreversible process

$$(\Delta S_{sys.} + \underbrace{\Delta S_{surr.}}_{0}) > 0$$

$\therefore \Delta S_{sys} > 0$ (For irr. adi. process)

(b) For solids and liquids :

$$(\Delta S)_{sys.} = \int_1^2 \frac{q_{rev}}{T} = \int_1^2 \frac{dU - W_{rev.}}{T}$$

$$= \int_1^2 \frac{nc_v dT + \cancel{PdV}}{T}$$

↙ ignored for solids & liquids

→ Solids & liquids are incompressible phases of substance, $dV \approx 0$

$$(\Delta S)_{sys.} = nC \ln \frac{T_2}{T_1}$$

* for solids & liquids $C_P \approx C_V = C$

(c) for phase transformation :

→ Phase change occurs at constant pressure & temperature, and is considered reversible if it occurs at its transition temperature (eg - M.P, B.P etc)

$$(\Delta S)_{sys.} = \int_1^2 \frac{q_{rev.}}{T_{transition}} = \frac{Q_P}{T_{trans.}} = \frac{\Delta H}{T_{trans.}}$$

- for fusion of solid,

$$(\Delta S)_{fusion} = \frac{\Delta H_{fusion}}{T_{M.P}}$$

- For vapourisation of a liquid,

$$(\Delta S)_{vap.} = \frac{\Delta H_{vap.}}{T_{B.P}}$$

Q. 2 mol of an ideal diatomic gas is isothermally expanded from 300K, 2 atm to 1 atm. Find $(\Delta S)_{sys.}$, $\Delta S_{surr.}$ and ΔS_T, If -

(a) Process is Reversible.

(b) Irr. and expansion occured against a constant external pressure of 1 atm. and 300 cal of heat is absorbed.

In (b) part, external pressure of 1 atm is not needed.Don't use it.

(c) Free expansion. (Adiabatic)

Soln: $T = 300K$, 2atm $\longrightarrow$ 1 atm, 2 mol

(a) Rev. Isothermal process:

$$\Delta S_{system} = nR \ln \frac{P_1}{P_2} = 2 \times R \times \ln \frac{2}{1}$$

$$= 2 \times 2 \times \ln 2$$

$$= 4 \ln 2 \text{ cal/K}$$

$\Delta S_T = 0$ (For reversible process)

$$(\Delta S)_{system} + (\Delta S)_{surr.} = 0$$

$$(\Delta S)_{surr.} = -4 \ln 2 \text{ cal/K}$$ Ans.

(b) Irr. Isothermal system.

$$\Delta S_{sys.} = nR \ln \frac{P_1}{P_2} = 4 \ln 2 \text{ cal/K}$$

$$\Delta S_{surr.} = \frac{Q_{surr.}}{T_{surr.}} = -\frac{300}{300} = -1 \text{ cal/K}$$

$$\Delta S_{Total} = 4 \ln 2 - 1$$

$$= 4 \times 0.7 - 1$$

$$= 1.8 \text{ cal/K}$$ Ans.

Imp (c) Free expansion or expansion against vaccum.

$W = 0, \quad q = 0, \quad \Delta U = 0$

$\Rightarrow \quad U = f(T)$.

$T \rightarrow$ constant

$$(\Delta S)_{system} = nR \ln \frac{P_1}{P_2} = 4 \ln 2 \text{ cal/K}$$

$$(\Delta S)_{surr} = \frac{Q_{surr.}}{T} = 0$$

$$(\Delta S)_{Total} = 4 \ln 2 + 0$$
$$= 4 \times 0.7 + 0$$
$$= 2.8 \text{ cal/K}$$

$\rightarrow$ Free expansion is an irreversible process.

Q. 4g of liquid oxygen (a) is converted into solid O_2 at its freezing temperature of 80K. Find entropy change of system surrounding, total if enthalpy of fusion of solid O_2 is 100 cal/mol and temp. of surrounding is 300 K.

Soln: $O_2(l) \rightarrow O_2(s), \quad \Delta H = -1000 \text{ cal/mol}$

$$(\Delta S)_{system\ (4g)} = \frac{-1000}{80} \times \frac{4}{32} = -\frac{25}{16} \text{ cal/K}$$

$$(\Delta S)_{surr.} = \frac{(1000 \times 1/8) \text{ cal}}{300} = \frac{5}{12} \text{ cal/K}$$

$$\Delta S_T = \left(-\frac{25}{16} + \frac{5}{12}\right) \text{ cal/K}$$

Q 1 mol of an ideal triatomic (non-linear) gas is adiabatically expanded from 16 atm, 300 K to 1 atm against a constant external pressure of 1 atm. Find $\Delta s_{sys.}$, $\Delta s_{surr.}$ and Δs_{total}.

Soln: Irr. adi. process, $Q_{irr} = 0$

$\Delta U = W_{irr}$

$nC_v\Delta T$

$\Rightarrow 1 \times 3R (T_2 - 300)$

$= -1\left(\frac{1 \times R \times T_2}{1} - \frac{1 \times R \times 300}{16}\right)$

$\Rightarrow 3T_2 - 900 = -T_2 + \frac{300}{16}$

$4T_2 = 900 + \frac{300}{16}$

$\Rightarrow T_2 = 225 + 75/16$

$\Rightarrow T_2 = 229.7\,K \approx 230\,K$

$(\Delta S)_{sys.} = nC_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$

$= 1 \times 4R \ln \frac{230}{300} + 1 \times R \ln \frac{16}{1}$

$= 4R \ln 2 - 4R \ln 30/23 = +ve$

$(\Delta S)_{surr.} = Q/T = 0$

$(\Delta S)_{Total} = \Delta S_{sys} + 0$

$= 4R \ln 2 - 4R \ln 30/23$

* Entropy (S) : (state function)

Entropy is the measure of randomness or disorderedness of a system. Randomness is associated with molecules of a substance as the no. of possible arrangements or orientation in which molecules / atoms of substance can be in.

Higher are no. of possible arrangements / orientation greater will be the randomness & hence greater will be the entropy.

Mathematically, $dS = \frac{q_{rev}}{T} \Rightarrow \Delta S = \int_1^2 \frac{q_{rev.}}{T}$

V. Imp Comparison of entropy on the basis of randomness :

1. For a substance, m → molar

$(S_m)_{solid} < (S_m)_{liquid} << (S_m)_{gas}$

2. Two metal blocks (identical) at diff. temp.

$T \uparrow, S \uparrow$

3. Two identical ideal gas samples at same temperature

$V \uparrow, S \uparrow$

$P \uparrow, V \downarrow, S \downarrow$

4. Greater is the complexity (includes atomicity also) Greater will be Molar Entropy under identical conditions.

$(S_m)_{O_3} > (S_m)_{O_2} > (S_m)_O$

$(S_m)_{H_2} > (S_m)_H$

$(S_m)_{CH_3COOH(\ell)} > (S_m)_{HCOOH(\ell)}$

5. (S_m) crystalline form $<$ (S_m) amorphous form for a substance.

$\therefore$ (S_m) white phosphorus $>$ (S_m) Red $>$ (S_m) Black.

$\therefore$ (S_m) graphite $>$ (S_m) diamond

6. for a chemical reaction, if Δn_g = +ve then $\Delta S_{Reaction}$ = +ve

if Δn_g = -ve
ΔS_R = -ve

eg. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$, $\Delta n_g = 2-4 = -2$
ΔS_R = -ve

eg. $C(s) + 1/2\, O_2(g) \longrightarrow CO(g)$, $\Delta n_g = 1/2$
ΔS_R = +ve

7. Mixing of two gases (Non-reacting)

↓

ΔS = +ve
(Entropy ↑)

8. **Heavier isotope has slightly greater entropy than the lighter isotope**

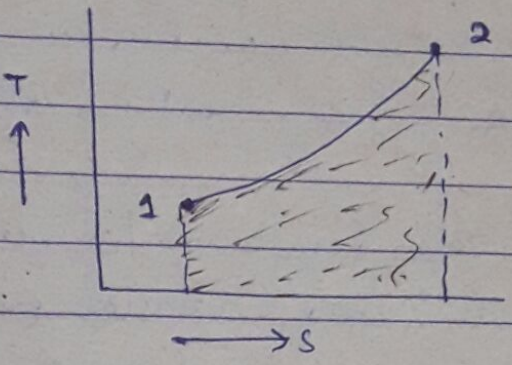
$(S_m)^{\circ}_{T_2} > (S_m)_{D_2} > (S_m)_{H_2}$

* Representation of Carnot Cycle in T-S diagram :

$$ds = \frac{q_{rev}}{T} \Rightarrow q_{rev} = Tds$$

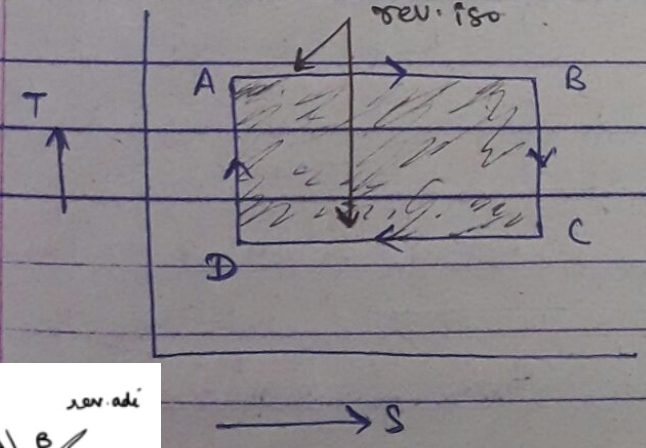
$$Q_{rev} = \int_1^2 Tds$$

→ Area - under T-S graph represents reversible heat transfer.

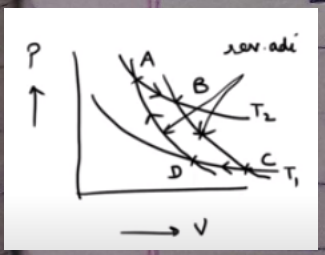


Carnot Cycle :

→ Two rev. isothermal and two rev. adiabatic process.



→ Rev. adiabatic process is called isentropic process.



In a cyclic process

$\Delta U = 0$

$Q_{rev} = -W_{rev}$

→ Area under graph represents magnitude of reversible heat transfer, or work done.

(magnitude of work done)

(*) Third law of Thermodynamics :-

Entropy of a perfectly crystalline element or compound approaches zero when temperature approaches absolute zero.

At absolute zero (or Zero Kelvin), there is only one preferred arrangement / orientation of substance and K.E becomes zero, then entropy will be zero.

gmp X **Note:** Some compounds / elements have non-zero entropy at 0 kelvin, called residual entropy.

eg. CO, NO, orthohydrogen → a nuclear isomer of molecular hydrogen based on spin of proton.

H_2 → orthohydrogen {parallel spin of protons}

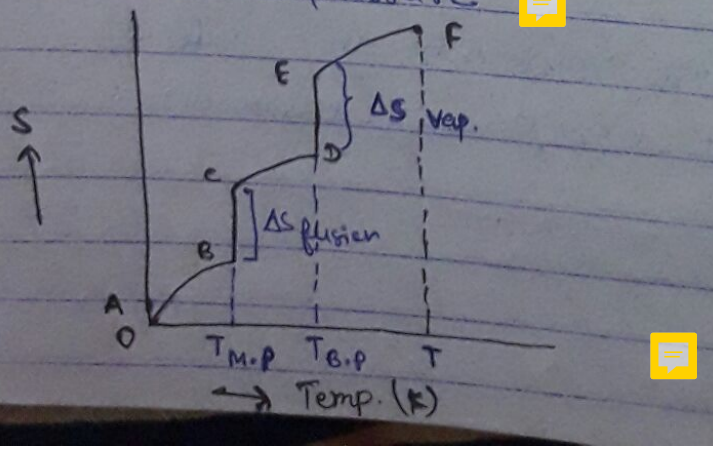
H_2 → Parahydrogen {Antiparallel spin of protons}

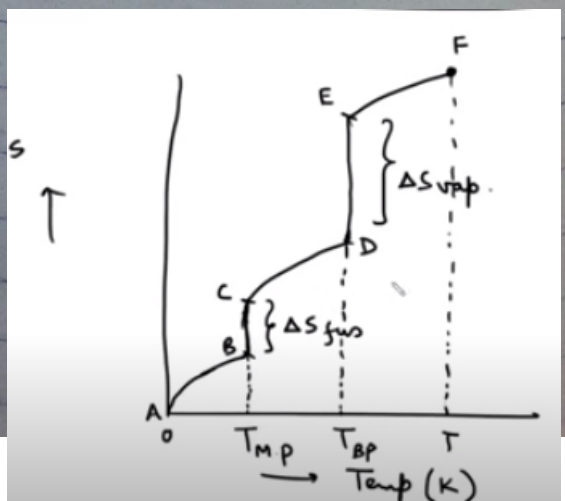
From 3rd law,

$$S_{zero\ 'K'} = 0$$

* Applications of 3rd law of Thermodynamic:

(1.) To determine absolute entropy of a substance at a temperature at constant pressure.





$$\Delta S_{Total} = S_T - \cancel{S_0}_{zero} = \Delta S_{AB} + \Delta S_{BC} + \Delta S_{CD} + \Delta S_{DE} + \Delta S_{EF}$$

by 3rd law

$$\Rightarrow S_T = \int_0^{T_{M.P}} \frac{nC_p(s)\,dT}{T} + \frac{\Delta H_{fus}}{T_{M.P}} + \int_{T_{M.P}}^{T_{B.P}} \frac{nC_p(l)\,dT}{T} + \frac{\Delta H_{vap.}}{T_{BP}} + \int_{T_{B.P}}^{T} \frac{nC_p(g)\,dT}{T}$$

(2) To determine Entropy of reaction (ΔS_r):

$$aA + bB \longrightarrow cC + dD, \quad \Delta S_r = \Sigma(S)_P - \Sigma(S)_R$$

$$\Rightarrow \Delta S_r = (c \times S_C + d \times S_D) - (a \times S_A + b \times S_B)$$

where S_A, S_B, S_C, S_D are molar entropies of A, B C and D respectively.

Q $N_{2(g)} + 3H_2(g) \longrightarrow 2NH_3(g)$, $(Sm)_{N_2} = 150$, $(Sm)_{H_2} = 100$, $(Sm)_{NH_3} = 200$ } J/K/mol

Find entropy of reaction.

Soln

$$\Delta S_r = 2 \times 200 - 150 - 3 \times 100$$
$$= 400 - 150 - 300$$
$$= -50 \text{ J/K/mol.}$$

Jee - All only

(*) Variation of entropy of reaction (ΔS_r) with pressure and temperature.

(a) At constant pressure :

$$aA + bB \longrightarrow cC + dD$$

Tem. T_1 $\quad aS_A \quad bS_B \quad cS_C \quad dS_D$

$$(\Delta S_r)_{T_1} = (c \times S_C + d \times S_D) - (aS_A + b \times S_B)$$

Tem. T_2 $\quad \left\{aS_A + \int_{T_1}^{T_2} \frac{a\,C_p(A)\,dT}{T}\right\} + \left\{bS_B + \int_{T_1}^{T_2} \frac{b\,C_p(B)\,dT}{T}\right\} \ldots$

$$(\Delta S_r)_{T_2} = \Sigma S_{product} - \Sigma S_{reactant}$$

$$= (cS_C + dS_D) - (aS_A + bS_B) + \int_{T_1}^{T_2} \frac{(\Delta C_p)_r \times dT}{T}$$

where, $(\Delta C_p)_r = \Sigma (C_p)_{prod.} - \Sigma (C_p)_{reactant.}$

$$= (c \times C_{pC} + d \times C_{pD}) - (a \times C_{pA} + b \times C_{pB})$$

$$\Rightarrow \boxed{(\Delta S_r)_{T_2} = (\Delta S_r)_{T_1} + (\Delta C_p)_r \ln \frac{T_2}{T_1}}$$

If $(\Delta C_p)_r = 0$ for given rxn. then

$$(\Delta S_r)_{T_2} = (\Delta S_r)_{T_1}$$

entropy of rxn. does not change with temperature

(b) At constant tempesature : (Pressure varied)

$$\Delta S = n C_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

$$aA + bB \longrightarrow cC$$

Pressure P_1 : $aS_A \quad bS_B \quad cS_C \quad , (\Delta S_r)_{T_1} = cS_C - aS_A - bS_B$

Pressure P_2 : $\left\{ aS_A + aR \ln \frac{P_1}{P_2} \right\} , \left\{ bS_B + bR \ln \frac{P_1}{P_2} \right\} \left\{ cS_c + cR \ln \frac{P_1}{P_2} \right\}$

new entropies at pressure P2

$$(\Delta S_r)_{P_2} = (cS_c - aS_A - bS_B) + (cR - aR - bR) \ln \frac{P_1}{P_2}$$

$$\boxed{(\Delta S_r)_{P_2} = (\Delta S_r)_{P_1} + \Delta n_g \times R \ln \frac{P_1}{P_2}}$$

Δn_g = Sum of st. coeff of all gas products − Sum of st. coeff of all gas reactants

* We have ignored entropy of solid & liquid in comparison to gases.

Q. (a) Calculate $\Delta S^\circ (300K)$ for the rxn.

$N_2(g) + 2O_2(g) \rightarrow 2NO_2(g)$. $(S_m)_{300K, N_2} = 160$, $(S_m)_{\gg, O_2} = 190$, $(S_m)_{\gg, NO_2} = 240$ J/K/mol

(b) Find $\Delta S^\circ (400K)$ for above rxn. if $(C_p)_{N_2} = 29$, $(C_p)_{O_2} = 29$, $(C_p)_{NO_2} = 37$.

Sol^n : (a) $(\Delta S_r)_{300K} = 2\times240 - 160 - 2\times190$

$= 480 - 160 - 380$

$= 320 - 380$

$= -60\ J/K/mol$

(b) $(\Delta S_r)_{400K} = (\Delta S_r)_{300K} + (\Delta C_p)_r \ln \frac{400}{300}$

$= -60 + (2\times37 - 29 - 2\times29) \times \ln \frac{4}{3}$

$= -60 + (6-13) \ln 4/3$

$= (-60 - 13 \ln 4/3)\ J/K/mol.$ Ans.

(*) Gibb's function / Gibb's Energy / Gibb's free Energy : (State function) (G)

→ Change in Gibb's energy at constant pressure and temp. establishes an alternate criterion of spontaneity or feasibility of a process or reaction.

→ It is a thermodynamic property of system.

→ It is defined as

$G = H - TS$

$dG = dH - TdS - SdT^{\to 0}$

At const pressure & temperature, $dT = 0$, $dH = q_p$

{ considering no non-pv work }

$$(dG)_{T,P \text{ for system}} = (dH)_{sys.} - T(dS)_{sys.}$$

$$= (q_P)_{sys.} - T(dS)_{sys.}$$

$$(dS)_{surr} = \frac{-(q_P)}{T} \Rightarrow q_P = -T(dS)_{surr}$$

$$(dG)_{T,P,\,system} = -T(dS)_{surr.} - T(dS)_{sys.}$$

$$(dG)_{T,P\ system} = -T(dS)_{total}$$

OR [IIT - Main]

$$\boxed{(\Delta G)_{T,P,\,system} = -T(\Delta S)_{total}} \quad \text{---(i)}$$

If $\Delta S_{total} = +ve$, $(\Delta G)_{T,P} = -ve$, Spontaneous process OR Irr. process.

If $\Delta S_T = 0$, $(\Delta G)_{T,P} = 0$, Reversible process or Eq^m. state.

If $\Delta S_T = -ve$, $(\Delta G)_{T,P} = +ve$, Non-spontaneous process or non-occuring process.

↑ an alternate criterion of spontanity.

$$\boxed{(\Delta G)_{T,P\ system} = \Delta H_{sys} - T\Delta S_{sys.}}$$

Q. $2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$, $\Delta H_r = -50\ KJ/mol$

$(Sm)_{NO} = 120$, $(Sm)_{O_2} = 80$, $(Sm)_{NO_2} = 150$ } J/K/mol at 300 K.

Comment whether the above rxn. is spontaneous or non-spontaneous.

Soln: $\Delta S_r = 2 \times 150 - 2 \times 120 - 80$

$= 300 - 240 - 80$

$= -20\ J/K/mol$

$(\Delta G)_{T,P} = \Delta H - T\Delta S$

$= -50 - \frac{300 \times (-20)}{1000}$

$= -50 + 6$

$= -44\ KJ/mol$ } -ve indicates that rxn is spontaneous under given conditions.

V. Imp (*) Physical Significance of ΔG:

For a process or reaction

$G = H - Ts$

$dG = dH - Tds - SdT$

$dG = d(U + PV) - Tds - SdT$

$dG = dU + pdv + Vdp - Tds - SdT$

$dG = (q_{rev} + W_{rev}) + PdV + Vdp - Tds - SdT$

$= Tds + (W_{PV})_{rev} + (W_{non-PV})_{rev} + PdV + VdV - Tds - SdT$

$= -pdV + (W_{non-PV})_{rev} + pdV + Vdp - SdT$

$\{(dG)_{T,P} = (W_{non-PV})_{rev}\}$

→ Change (or decrease) in Gibb's function in a reaction or process at constant P and T is the measure of max. non - PV work that can be obtained from system.

→ Non - PV work is called useful work or free available energy.

* Electrical work (eg. obtained in a Galvanic cell) is an example of non - PV work.

⇒ Electrical work energy from a Galvanic cell = $(\Delta G)_{T,P}$

$= -nFE_{cell}$.

Note : (Memorise)

① For a reaction,

$aA + bB \longrightarrow cC + dD$,

$$\Delta G^\circ_r = \Sigma(\Delta G^\circ_f)_{product} - \Sigma(\Delta G^\circ_f)_{reactants}$$

$$(\Delta G^\circ_r) = \{c \times (\Delta G^\circ_f)_C + d \times (\Delta G^\circ_f)_D\} - \{a \times (\Delta G^\circ_f)_A + b \times (\Delta G^\circ_f)_B\}$$

→ ΔG°_f (Standard Gibb's energy of formation of all the elements in standard state (generally most stable state) is assigned zero value)

eg. $\Delta G^\circ_f (O_2)_g = 0$

$\Delta G^\circ_f (Br_2)(l) = 0$

$\Delta G^\circ_f (Fe(s)) = 0$

} at 25°C or 298 K.

* What is free in Gibb's free energy :

eg. Combustion of butane ($C_4H_{10}(g)$)

$$C_4H_{10}(g) + 6.5O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$$

$(\Delta H_r)_{300K} = -100\ kJ/mol$

$\Delta n_g = 4 - 7.5$
$= -3.5$

$(\Delta S_r)_{300K} = -40\ J/K/mol$

$(\Delta S)_{surr} = \frac{100 * 10^3}{300} = 333.3\ J/K/mol$ { when entire heat is lost to surrounding.

$\Delta H_{300K} = -100\ kJ/mol$
$\Delta S_{sys} = -40\ J/K/mol$
$(\Delta S)_{surr} = 333.3\ J/K/mol$

⇒ If we want just process (reaction) to occur spontaneously

$\Delta S_T \geq 0$

$\Delta S_{sys} + \Delta S_{surr} \geq 0$

⇒ $\Delta S_{surr} \geq 40\ J/K$

this is the minimum entropy change that must take place for rxn. to occur.

$\{(\Delta S)_{surr}\}_{min} = \frac{\Delta H_{min.\ surrounding}}{300}$

⇒ $(\Delta H)_{min.,\ surr.} = \frac{300 \times 40}{1000}\ kJ/mol$

$= 12\ kJ/mol$

* A minimum of 12 KJ heat (out 100 KJ liberated) should be lost to surrounding (or should be given to nature as bribe).

$$(\Delta G) = \Delta H - T\Delta S$$

$$= -100 - \frac{300 \times (-40)}{1000}$$

$$= -100 + 12$$

$$= -88 \text{ KJ/mol (free available energy)}$$

OR

useful work.

$\eta_{cell} = \frac{|\Delta G|}{|\Delta H|} \times 100$ = energ or u

$= \frac{88}{100} \times 100 = 88\%$ Ans

efficiency of fuel cell.

* Variation of Gibb's function (G) for a substance with pressure and temperature :-

$$G = H - Ts$$

$$dG = dH - Tds - SdT$$

$$dG = d(U + PV) - Tds - SdT$$

$$dG = dU + PdV + VdP - Tds - SdT$$

$$dG = (q_{rev} + w_{rev}) + PdV + VdP - Tds - SdT$$

$$dG = Tds + (-PdV) + PdV + VdV - Tds - SdT$$ { considering no non-PV work }

$$\Rightarrow \boxed{dG = VdP - SdT}$$

$$\left(\frac{\partial G}{\partial P}\right)_T = V \qquad \left(\frac{\partial G}{\partial P}\right)_P = -S$$

$$\Delta G = \int_1^2 VdP - \int_1^2 SdT$$

(a) For Solids and liquids (at constant temperature)

$$dG = Vdp - SdT \quad (SdT \to 0)$$

$$\Delta G = \int_1^2 V dP$$

$$\boxed{\Delta G = V(P_2 - P_1)}$$

V of solids & liquids are assumed constant.

(b) For an ideal gas (at constant temperature)

$$\int_1^2 dG = \int_1^2 V dP$$

$$\int_1^2 dG = \int_1^2 \frac{nRT}{P} dP$$

$$\Rightarrow \boxed{\Delta G = nRT \ln \frac{P_2}{P_1}}$$

Q. Pressure of 1 mol of liquid water is increased from 1 atm to 10 atm, find change in Gibb's energy.

Soln: $V_{H_2O(l)} = 18 ml$

$$\Delta G = V(P_2 - P_1)$$
$$= 18 \times 10^{-6} m^3 \times (10-1) \times 1.013 \times 10^5 Pa$$
$$= 18 \times 9 \times 0.1 \times 1.013 \text{ Joule}$$
$$= 16.4106 \text{ Joule}$$

Q. 8 g of methane gas is isobarically heated from 300 K to 400 K, find change in Gibb's energy if $(S_m)_{CH_4} = (10 + 10^{-2} T)$ J/K/mol

Sol^{n} : $dG = \cancel{Vdp} - SdT$

$$(\Delta G)_{1\,mol} = -\int_{300}^{400} (10 + 10^{-2})\, dT$$

ΔG for 8g of CH_4 $= \frac{1}{2} \times \left[-10(400-300) - \frac{10^{-2}}{2}(400^2 - 300^2) \right]$

$$= \frac{1}{2}\left[-1000 - 50 \times 7 \right]$$

$$= \frac{-1350}{2} = -675 \text{ Joule.}$$

Imp. (*) Free energy and Equilibrium :

Consider the equilibrium system,

$$aA_{(g)} + bB_{(g)} \rightleftharpoons cC_{(g)} + dD_{(g)}$$

$dG = Vdp - SdT$ { for a substance }

At constant temperature, $dT = 0$

$dG = Vdp$,

Treating gases to be ideal, $PV = nRT$

$$\Delta G = \int_1^2 \frac{nRT}{P}\, dP = nRT \ln \frac{P_2}{P_1}$$

For substance (or gas) A,

$$G_2 - G_1 = aRT \ln \frac{P_2}{P_1}$$

$$G_A = G_A^{\circ} + aRT \ln \frac{P_A}{P_A^{\circ}}$$

$P_A^{\circ} = 1$ bar (standard pressure)

$\Rightarrow G_A = G_A^{\circ} + RT \ln P_A^{a}$

$\Rightarrow G_B = G_B^{\circ} + RT \ln P_B^{b}$

$\Rightarrow G_C = G_C^{\circ} + RT \ln P_C^{c}$

$G_D = G_D^{\circ} + RT \ln P_d^{d}$

$\Delta G_r = \Sigma (G)_{product} - \Sigma (G)_{reactants}$

$\Delta G_r = \Delta G_r^\circ + RT \ln \left(\frac{P_C^c \times P_D^d}{P_A^a + P_B^b} \right)$

$\Rightarrow$ $\boxed{\Delta G_r = \Delta G_r^\circ + RT \ln_{10} Q_p}$ —— (1)

where ΔG_r = Gibb's energy of reaction

ΔG_r° = Standard Gibb's energy of reaction.

Q_p = Reaction quotient.

At eqm. state, $\Delta G_r = 0$ and $Q = K_{eq}$.

$0 = \Delta G_r^\circ + RT \ln K_{eq}$

$\Rightarrow$ $\Delta G_r^\circ = -RT \ln K_{eq}$

$\boxed{\Delta G_r^\circ = -2.303\, RT \log_{10} (K_{eq})}$

$= \Delta H_r^\circ - T \Delta S_r^\circ$

$\Rightarrow \log_{10} K_p = \frac{-\Delta H_r^\circ}{2.303RT} + \frac{\Delta S_r^\circ}{2.303R}$

* Remember, At eqm. state,
$\Delta G_r = 0$ and ΔG_r° need not be zero

* ΔG_r° will be zero only when eqm exist under standard conditions. $K_p = 1$

$\Delta G_r^\circ = -2.303RT \log K_p$

if $K_p = 1 \Rightarrow \Delta G_r^\circ = 0$

Q. For the equilibrium

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

$$\ln K_p = \frac{10^4}{T} = -10$$

Find ΔH° and ΔS° and ΔG° of reaction at 300K.

Sol^n:

$$\Delta G^\circ = -RT \ln K_p = \Delta H^\circ - T\Delta S^\circ$$

$$\Rightarrow \ln K_p = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

Comparing,

$$-\frac{\Delta H^\circ}{R} = 10^4 \Rightarrow \Delta H^\circ = -10^4 R$$

$$\frac{\Delta S^\circ}{R} = -10 \Rightarrow \Delta S^\circ = -10R$$

$$(\Delta G^\circ)_{300} = -10^4 R - 300 \times (-10R)$$
$$= -7000R$$

(*) Effect of temperature on Spontaneity of a Reaction :-

$$\Delta G_{T,P} = \Delta H - T\Delta S$$

* Exothermic ness and increase in entropy of system favour the spontaneity.

	ΔH	ΔS	ΔG	
a)	-ve	+ve	-ve (always)	{Rkn is spontaneous under all temp.}
b)	+ve	-ve	+ve always	{Rkn is non-spontaneous}
c)	+ve	+ve	[-ve at high temp (or spont. at high temp) +ve at low tem (or non-sp. at low temp)	

eg. melting of iron

⇒ $-ve \cdot -ve$ [$-ve$ at low temp. { spon. at low temp. / $+ve$ at high temp. { non-spon. at high temp.

eg. adsorption of gas on solid surface

V.V. Imp

Q. Predict spontaneity of the following phase change and comment on sign of ΔG.

(i) $H_2O(s, 1atm, 273K) \longrightarrow H_2O(l, 1atm, 273K)$

(ii) $H_2O(l, 1atm, 373K) \longrightarrow H_2O(g, 1atm, 373K)$

(iii) $H_2O(l, 2atm, 373K) \longrightarrow H_2O(g, 2atm, 373K)$

(iv) $H_2O(l, 0.8atm, 373K) \longrightarrow H_2O(g, 0.8atm, 373K)$

Soln: Hint: P↑, B.P ↑ for a liquid

(1.) $\Delta G = 0$ (Reversible) (2) $\Delta G = 0$ (Reversible)

(3) $\Delta G = +ve$ (Non-spontaneous) (4) $\Delta G = -ve$ (spontaneous)

as B.P increased **as B.P is decreased**

Q. Calculate ΔG for phase change

$H_2O(l, 2atm, 373K) \longrightarrow H_2O(g, 2atm, 373K)$

Soln:

$$H_2O(l, 2atm), 37\cancel{3}K) \xrightarrow{\Delta G} H_2O(g, 2atm, 373K)$$

$$\downarrow \Delta G_1 = V(P_2 - P_1) \qquad \uparrow \Delta G_3 = nRT \ln P_2/P_1$$

$$H_2O(l, 1atm, 373K) \xrightarrow[\Delta G_2 = 0]{} H_2O(g, 1atm, 373K)$$

$$\Delta G = \Delta G_1 + \Delta G_2 + \Delta G_3$$

$$= 18 \times 10^{-6}(1-2) \times 1.013 \times 10^5 + 0 + 1 \times \frac{25}{3} \times 373 \ln \frac{2}{1}$$

$$= -1.8 \text{ Joule} + 25/3 \times 273 \times 2.3 \times 0.3 \text{ Joule}$$

$$= +ve$$